Graduate School in Applied Mathematics: Guide for Students and New Faculty

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Preface

This book is about the advisor-student collaboration. Our audience consists of students considering or working towards a PhD in applied mathematics at all stages of their studies.

Many students enter graduate school without a clear understanding of their responsibilities, the milestones that define progress, or what they can reasonably expect from their advisor. Likewise, new faculty often receive little to no training in how to mentor PhD students. They know their own graduate experience, which may or may not serve as a useful model, but stepping into the advisor role comes with its own challenges.

Our aim is to clarify these roles and expectations. Think of this book as a "consumer guide" to graduate school: a resource for students navigating the path from admission to graduation and employment, and for advisors—especially junior faculty—learning how to guide students through the process. While the graduate student experience shares commonalities across disciplines, the focus here is specifically on applied mathematics in the United States. This brings in features like connections with national laboratories, disciplinary culture, and professional societies that may not apply as directly in pure mathematics or applied mathematics outside the U.S.

The book traces the PhD journey from both perspectives, focusing on milestones and responsibilities: from selecting an advisor, to research and thesis writing, to landing that first job. Along the way, we include several *vignettes* that reinforce the ideas in this book. Some of the vignettes were contributed anonymously by graduate students and professional mathematicians in academia, industry, and government in response to our solicitations. We took others from the *Working Life* section (see § A.2) of the journal Science and the citations are within the vignettes. These stories—some lighthearted, others sobering, show the real challenges and triumphs of the process. Our hope is that they not only demystify graduate school, but also reassure readers that they are not alone in their experiences.

Ultimately, this book is meant to ease anxieties, provide practical guidance, and highlight best practices for both students and advisors. Success in graduate school comes from a combination of preparation, communication, and perseverance. By understanding the expectations on both sides, students and advisors alike can navigate the PhD process more effectively.

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Chapter 1
Graduate School

1.1 ▪ The Process

Although the details may vary widely from university to university, the process of entering and exiting a graduate program (successfully with a PhD) in applied mathematics has some core common features; an admission process, coursework, passing qualifying exams, the selection of an advisor, and a proposal and final PhD defense that rely on the student's original research. Some of those events may happen in a slightly different order but the beginning and end are the same. The bulk of this book will focus on the expectations and responsibilities that exist between the advisor and the advisee in order to provide guidance for both people. In this chapter though, we give a brief overview of the graduate school process and those common components mentioned above. More details are given as needed throughout the book.

1.1.1 ▪ Admission

As a student, you should carefully check the admission requirements for the schools you are interested in, but application packets for graduate school typically consist of undergraduate transcripts and possibly GRE scores (some universities require the subject test). It may also require a curriculum vita (much like a longer, detailed resume) that would describe any research experiences, special coursework, internships, math competitions, presentations, or even service you may have done that would highlight your skills. Critical to any application are several letters of recommendation. During their undergraduate studies, a student needs to identify people (professors or possibly advisors/bosses) that they trust to communicate just how innovative and hard-working they are. Transcripts will show that a student can get good grades, but a letter that speaks to a student's curiosity in problem solving, thirst for knowledge, integrity, work ethic, maturity, and perseverance are just as important, maybe even more important. Note that students do not get to see these letters—usually they are submitted directly from the letter writer to the admissions office.

Finally, a statement of purpose (SOP) that describes why you want to go to graduate school at that particular university is the chance to put a personal spin on the big picture. The SOP is your opportunity to sell yourself beyond your transcript and CV. Your SOP should be well-written, thoroughly checked for grammar and may include the following;

- What inspired you to pursue advanced studies in mathematics.
- What draws you specifically to that math department.
- It can help to identify specific faculty members or research centers on campus that you might want to work with. This shows you took the time to learn what the department is like.
- Description of how getting a PhD in applied mathematics connects to future goals.
- Any explanations of any anomalies in your CV or transcript (but be careful not to overshare).
- Special experiences such as REUs (Research Experiences for Undergraduates), internships, competitions, special projects, leadership roles, or teaching experiences.
- Most importantly, describe what makes you unique.

Your SOP should not include blunders like using a different university's name or saying that you want to work with someone who is actually a retired faculty member. You should be applying to schools that are a good fit with your interests. Know what research is being done. For example, you shouldn't describe your passion for Lie Algebras if you are applying to a department full of people who work with numerical partial differential equations.

Writing an SOP can be challenging for some people. It doesn't often feel natural to sell ourselves and talk about our achievements, but it is necessary here. Graduate admissions, teaching assistantships, and research fellowships are competitive, so the SOP should be taken seriously so you stand out. Start early and get feedback on it, perhaps from one of your undergraduate professors.

1.1.2 ▪ Undergraduate Research

As an undergraduate you have a chance to impress the faculty. If you have the opportunity for undergraduate research, exploit that opportunity. Doing well will get you very effective supporting letters for application to graduate school.

There are many entry points to get started on research as an undergraduate. One way is to participate in an REU as mentioned above. REUs usually take place over the summer for about ten weeks. The application process starts around February. The National Science Foundation maintains a list of funded REUs `https://www.nsf.gov/funding/initiatives/reu/search` and you can filter by topic and location. Most REUs provide a stipend, may include housing and meals, and often include additional professional development opportunities. Student participants usually work in teams or pairs under the guidance of a research mentor (much like the graduate school experience). REUs are an opportunity to learn if you enjoy research and decide if graduate school would be a good fit for you. If you are an undergraduate reading this book, we encourage you to look at what possibilities are out there. In particular, if you are already considering going to a certain graduate school, then see what they have to offer.

Another way to get involved in research as an undergraduate is to seek out an advisor. Be mindful that just having an interest will not be enough and many faculty will not take on an undergraduate research student unless that student has impressed them already in a class. The junior author of this book has worked with numerous undergraduate researchers with primarily positive experiences including co-authoring papers with them, taking them to conferences, watching them win awards, and seeing them go off to graduate school. However, even as an undergraduate, you have obligations in a research project. Juggling courses and a research project can be challenging so it is important not to take on too much.

One frustrated faculty member says:

> **Vignette:** *I've definitely had my share of undergrad research students who underperformed. I will always move forward by having students sign a contract where we both delineate what we can expect from one another. It can be helpful to have a student help write part of the contract so they have ownership for the contract.*

Having said that, undergraduate research can be nothing short of a life-changing experience that sparks a deeper excitement for mathematics and positions you well for your future. If you succeed you will make your research advisor very happy.

> **Vignette:** *I recruited one of my best undergraduate research assistants by asking our honors program coordinator for some names. Within five minutes of meeting him, I hired him. He got up to speed on optimization and a very challenging application very rapidly and was contributing to a funded project within a semester. He was with the project for only eighteen months and got his name on two journal publications and two technical reports from that time. He went on to a top graduate program and is now a senior person at a national laboratory.*
> *I can take credit for staying out of his way and not much else.*

Once you have everything submitted, it may seem like your application packet just goes into an abyss as you wait to hear from schools and hope for offers of financial support (possibly as a teaching assistant or a research fellow). Behind the scenes, a committee of faculty members is reading over applications, possibly scoring and ranking them, and discussing the potential success of each person. We should note—at most universities faculty are required to do some sort of service within their department. Serving on the graduate admissions committee is one of those possibilities. As a faculty member, serving on this committee may be one of the most important roles you could take on. Moreover, it can be beneficial to have a first look at applicants and see who may be a good fit for either your research program or possibly a colleague's. The acceptance criteria certainly differ from institution to institution. Whether or not a student will be successful academically, has the right background and coursework, and has a good research fit with the faculty all play an important role. However, decisions may also depend on the entire pool of applicants or how many teaching positions or funding opportunities are available. Ultimately, it is up to the committee.

1.1.3 ▪ Coursework

Upon arriving at graduate school, there is usually some form of an orientation and then a selection of coursework. Some students come in to graduate school already knowing what they want to study while others may want to take various classes to see what excites them. Exploring for a bit is good, but keeping the overall time line in place is important in order to sufficiently focus on the relevant coursework for research. In either case, everyone starts out with coursework—usually three classes per semester and possibly a seminar. Courses often consist of some core classes that comprise a solid foundation in upper level mathematics and then some special topics courses that prepare for a specific research pathway. For applied mathematics, this may include taking some engineering courses for example. Understanding the qualifying exam policy at your university will also dictate some of the courses you take. You will need to focus seriously on all your courses, but especially on those. Students tend to finish up their required coursework by their third (possibly fourth) year and then focus solely on the thesis. Keep in mind there is an urgency in finishing coursework so that time can be spent on research, which is at the heart of getting a PhD .

As a student or an advisor—you have to know what the requirements are. For example, some schools may require a foreign language course and others may require some coursework outside the department. This will be unique to your university and it is both of your jobs to be sure those requirements are met.

1.1.4 ▪ Qualifying Exams

Probably one of the most challenging aspects of graduate school is the qualifying exams, which again will vary widely across institutions. They will likely cover multiple courses usually from a specific list. They may be given at specific times of the year, be written or oral or a combination of both, or need to be completed by a certain time frame (for example after two years). The exams really are a first rite of passage in terms of getting the PhD . In some departments, students are not expected to begin research until they have passed them or perhaps faculty will not spend time on a student who has not yet passed them. As a student, understanding the culture of your department in this regard is important. As an advisor, you should understand that you shouldn't base the policies or practices at your institution on what your experiences were like when you were a graduate student someplace else. You need to ask your department chair and new colleagues about how the exams are administered and assessed and how they fit in to the overall framework for students and certainly ask to see some old exams if you are asked to write one.

As a graduate student, you need to understand the nature of how the exams are given—what is the format, how long will it take, what is the expected content, how are they graded, who grades them, when will you get the results, how many chances do you get to pass them? Find out as much as possible so you can have no doubt you are ready. No matter what the nature of qualifying exams are, it is vital that students take them seriously and prepare. Nothing can compare to the feeling of passing those! Doing well on your qualifying exams will help you attract a good advisor. Barely passing will make it hard to find an advisor at all.

As a faculty member, you should be supportive during this time—giving your advisee an opportunity to study.

1.1.5 ▪ Find an Advisor

What constitutes a successful working dynamic between an advisor and advisee is the main focus of this book so we only briefly discuss the notion of choosing an advisor here. It is a theme that is integrated throughout the following chapters. However, choosing an advisor is one of the most important decisions you can make as a graduate student. Likewise—as a faculty member, your research agenda depends on the capabilities and deliverables from your advisees. So, this is a match not to be taken lightly and both people need to give serious consideration before committing. In this sense, students may have an advantage in that they can usually do some digging about a faculty member. Look at their CV and home page. Ask questions like the ones outlined in Section 2.2. For a new professor who has never had a graduate student before, advising may seem exciting but also scary and overwhelming. This book offers some guidelines for successfully launching this aspect of an applied mathematician's academic career. In particular, Chapter 3 is dedicated to the advisor, starting out with recruiting a PhD student and ending with the student's job search.

1.1.6 ▪ Do Research and Write a Thesis

Ultimately, a student-advisor pair should produce accomplishments in mathematical research. Proposing a research topic, conducting research, and writing and defending a PhD thesis are really at the heart of what graduate school is all about—and all strongly connected back to the advisor-advisee relationship. In reality, the details of proposal and defense processes themselves are very much dependent on the institution—and new faculty need to be sure they understand those processes as they may differ wildly from their own experiences.

Once you and your advisor have agreed on a thesis project, it's time to build your thesis committee. This is a small (usually four or five) person group that signs the documents for your oral exams and approves your plan of study. There are many ways to staff the committee and you may or may not have much input. If the project is highly collaborative, you may be required to have several of the collaborators on your committee and find that your committee is fully staffed. Your advisor may ask you for your ideas and, if she/he does, don't try to put an "easy" person on your committee. You benefit from committee members who are engaged and give you useful feedback, even if that feedback means extra work for you. You also benefit from committee members from other disciplines, who can write very effective letters of recommendation for you.

The proposal is a description of the research project that clearly defines the research question and the approach that will be taken. It is meant to highlight the significance and originality of the work. At some universities, the proposal takes place after measurable progress has been made (for example in the 4th year) while at other places, the proposal really is a mechanism to seek approval by a committee that the research direction is valid and sound (for example at the beginning of the 3rd year after qualifying exams have been passed). The proposal may consist of a presentation to pre-selected committee members and may or may not require a written document given to the committee a week or so before the talk. There may be official paperwork that needs to be signed and submitted to an office on campus that documents the proposal was successful.

Most universities say a thesis must represent "significant and original work". Your thesis will be a scholarly document with a thorough literature review, original results, and a clear account of what you did and what your contributions to the field are. Moreover, since you are an applied mathematician, your thesis should clearly and directly say why people should care about your work. Your advisor should give you direction and support, but should not write your thesis for you. If you aspire to a research career, it's also important that you publish some of your results before graduation.

The procedures for a PhD defense also vary but the core requirement is that a student should have written a PhD thesis which has been carefully vetted by their advisor and consists of published or publishable work. Students should look at examples of theses to get a better understanding of how they are structured and what they contain. The thesis must be given to the committee members well in advance of an oral defense (the standard rule is two weeks) so that they have enough time to absorb the details and formulate questions and comments. Both the proposal and defense can be stressful for a student, but really it is a chance to get feedback and move in an appropriate direction with proper guidance. A student should expect to give an overview of their work to a broad audience (their committee plus members of their department) and then spend time with only their PhD committee members fielding specific questions about the validity and relevance of their work and future directions.

At every step of the process, the advisor should be advocating for the student and not letting the student move forward unless absolutely ready. This is the most important role of the advisor. In the following chapters, we will discuss in depth the various roles and expectations of the advisor and the advisee so that the process described above can not only end in success but go smoothly while being enjoyable for both people.

1.2 ▪ About Using Large Language Models

At the time this book is being written, large language models (LLMs) like ChatGPT [42] are emerging rapidly. Universities are developing policies about how they can and cannot be used. A student's ability to communicate, especially in writing, is critical at each step in the graduate school process–starting with the admissions letter all the way through the employment cover letter. Writing skills are essential and should be honed so that you are not relying on tools to write for you. There can be no doubt that artificial intelligence is a powerful tool in brainstorming, generating code, and designing research projects but it should always be used appropriately. We touch on these issues throughout the book when necessary. One thing you must attend to when you use an LLM to assist with programming is that the LLM may well have taken proprietary code without permission and you could be liable for using that code. Using an LLM to write parts of a paper or your thesis exposes you to the risk of committing plagiarism. You can get in serious trouble for plagiarism and cannot hide behind an LLM. The authors happily used an ChatGPT to proofread this book with the prompt *proofread for spelling, grammar, puncutation*. Your success with an LLM will depend on how well you structure the prompt.

1.3 ▪ The Advisor-Advisee Relationship

Students usually arrive in graduate school with very little knowledge about how the business relationship with an advisor works. One objective of this book is to address that. The starting point is when the student and advisor agree to work together, ideally after some thought. The student and advisor should consider more than the scientific project. For example, if one of them expects to work constantly and the other has strong views on work-life balance, that is trouble in the making and will create unnecessary stress for both. Advisors should be clear on their expectations and students should try to find advisors with working styles similar to their own. Section 2.2 has more, many more, details on this.

Both parties should keep in mind that this is a business relationship, not a personal one, and behave professionally when working together. Advisors should keep their opinions, if any, on a student's personal life, to themselves. Advisors are in a position of power and must take care not to abuse that power. Advisors should not offer advice on non-professional matters to your students. Students should also not expect their advisor to give counseling for personal problems or function as a family member. Every university has a counseling center where the staff is trained to deal with personal issues. While a student should expect an advisor to be understanding when the student has personal problems, the advisor is not the person to solve them.

Throughout the rest of this book, the important milestones on an applied math PhD student's pathway are described from both the student and advisor's viewpoint, with an emphasis on roles and responsibilities. We encourage both the student and the advisor to read everything so that they are more familiar with the similarities and differences in those viewpoints and have a better understanding of the processes.

Part I

For the student

Chapter 2

The Student

2.1 ▪ The Student's Rights and Obligations

You and your advisor are colleagues and your advisor should treat you as an equal. A thesis project is a collaboration, not an assignment. This simple idea has consequences.

- Your advisor will expect you to show up for appointments prepared and on time. You have the right to expect the same of your advisor. An advisor who is not prepared for meetings with students is doing a poor job. This is a warning sign that you might want to find a different advisor. On the other hand, if you consistently fail to show up on time or unprepared, it's a warning sign for your advisor, and she/he would do well to get rid of you.

  Most advisors are prepared for meetings and you should expect these meetings to be productive. Meetings are good times to talk about more than your research. If you have questions about careers, internship opportunities, or specific issues that you are having related to graduate school, it's fine to bring them up.

- If you are part of a research group, you may have obligations to the team. Sometimes funding agencies want your advisor to appear with a presentation on the group's activities. Your funding and that of the team may be on the line. If your advisor asks for viewgraphs or a report on very short notice, you must help. You may also be asked to stand-in for one of your advisor's courses while your advisor is on travel. If you plan to teach and the course is at the graduate level, this is an opportunity for professional development. It's fair for your advisor to ask, but you should be able to say no. If your advisor's travel is related to your project and your funding, you should probably agree to do it if you have the time.

- It's sad that we have to say this, but we do. It is not your job to make coffee, arrange travel for your advisor, babysit, do yard work, or do any kind of non-academic/research work that an administrative assistant or a household helper should be doing.

  Many students bring coffee and snacks to their oral exams. This is not a requirement. This article [9] recommends against it. The authors of this book agree.

- It is not your advisor's job to make travel arrangments for you either. You should expect to be able to figure out your own travel arrangements to conferences and stick to a budget. Learning how to do these things is part of professional development.

- Many, but not all, advisors are on a first-name basis with their students. Does this matter to you?
- You are responsible for getting your share of the project work done. If your project is interdisciplinary many other people will be depending on your timely completion of your assignments. Your obligations to the entire group come with pressure. Can you thrive with this pressure?
- You are responsible for upholding integrity. This is a core value in all aspects of academia whether it is in your coursework or your research. This includes doing your own work, giving credit to other people's work, and showing respect for other students, faculty, and staff within your department, univeristy, and scientific community.

Your advisor is responsible for making sure that your work leads to a degree, a thesis, publications, and a job. An advisor who does not seem interested in these things is not doing his/her job properly. Most advisors care very much about your future and you should expect to do your part. Again, we suggest you also read the chapter in this book about the advisor's rights and obligations as well.

2.2 ▪ Choosing an Advisor

It's a good idea for a student to do some research on a potential advisor. One can start this work as one looks at graduate schools.

Look up a potential advisor's home page, Google Scholar page, or LinkedIn page. Read over their CV. Is she/he a successful advisor? One can often see advisee successes from the advisor's home page. Where do her/his advisees find jobs? It's a very good sign if a professor's home page not only features her/his PhD students, but shows pride in them and how their careers are going.

If there's no information about the advisees on a faculty home page, that is not a good sign.

You should expect an advisor to have an up-to-date web page with current information on funding, students, and publications. If a potential advisor in applied mathematics does not have a web page at all, that is a very bad sign.

You should be able to find answers to questions like this on a potential advisor's home page.

- Are they publishing papers and presenting at conferences? You want an advisor who is professionally active and visible in the community. You also need an advisor who can guide you in writing your own papers and making presentations.

  A well-known and well-respected advisor can be a significant asset when you are looking for your first job. However, such an advisor may have obligations to professional societies or be a director of an institute or center. You should ask if these obligations have an impact on the time an advisor can spend with students.

- Are they attracting funding? If you want a research assistantship (RA) or want to travel to conferences, this is very important. Funding is also an indicator of the person's visibility and impact.

- Do the potential advisor's students get jobs? Does the advisor help her/his students with their job search? You do not want an advisor who leaves you on your own when it's time to find your first job. Does your advisor have contacts that hire new PhDs ?

  Equally importantly, do a potential advisor's students get the kinds of jobs you want to get? If you aspire to a teaching career and most of a potential advisor's students get jobs in industry or national laboratories, that advisor is probably a poor fit for you.

  You are likely thinking about your career interests before you start applying to graduate school. Do you want a program that sends most of its graduates to four year teaching colleges? If you do, focus your applications on those programs. The logic is the same for careers in national laboratories, industry, and research universities. The sooner you decide on a career objective, the better your chances for picking a graduate program and an advisor that can get you there.

Once you are in a graduate program, talk to other students in the program about what potential advisors are like to work with. Think hard about how you want the interaction to go. You need to know:

- Does the advisor spend enough time with her/his students? If the advisor has many students, how does she/he allocate time for them?

Is the advisor paying attention to her/his students' progress? How much attention from an advisor do you want? You should want and expect weekly meetings, but not all advisors have the time for more frequent meetings than that.

Many senior advisors have significant travel commitments, which will limit the time spent with students. Can you work alone while your advisor is away? Will you be content with virtual meetings? Virtual meetings can work very well and in some cases are the only option.

A senior advisor may have many students and a large research program. This usually comes with funding for RAs and travel for students. If you pick a senior and well-funded advisor, you will be one student of many and may not get as much attention as you want. A junior advisor may not have the level of funding to support you as a full-time RA or take you to many conferences. However, if you are someone's only student, you should get lots of attention.

- Is the advisor open to student ideas? Does she/he encourage independent thinking? Some students want an advisor to map out the entire thesis project, others want something open-ended. You want an advisor who will manage your project the way you like.

- When you write a paper with your advisor, the ordering of authors may matter to a potential employer. While it is traditional in mathematics for authors to appear in alphabetical order, that is not the case in most of science and engineering. If you've done your share of the work, it would be nice for your name to be first.

- Does the advisor take students to conferences? Your professional visibility is important and presenting at conferences is the best way for a student to build a network and meet the people in the field. This is very important for your future career opportunities and is connected to an advisor's funding.

- Do the advisor's students find internships in industry and national laboratories? Do the students think the internships are productive and useful? This is important if you are not interested in an academic career and can also help finding post-docs that lead to academic careers. The advisor's personal connections are an important part of your success in finding internships.

- What kind of students get along with a potential advisor? Are you that kind of student? While personality is part of this, the practical side of an advisor's research program is more important. For example, if a potential advisor has funding with deadlines and milestones, the advisor must put pressure on students to meet those deadlines. Do you deal with pressure well? Can you contribute to frequent reports?

- Do the advisor's students feel that she/he cares about their progress? Does the advisor help with a job search? Is the advisor paying attention and working with the student on things like the CV, the cover letter for applications, and networking. Your advisor should not leave you on your own at this critical career stage.

- Does the advisor treat students well and fairly? Is the advisor a bully? By talking to other students you can avoid these kind of advisors and the problems of working with one or switching advisors late in your degree program. § 6.1 says much more about this issue.

- Do you seem like a good match? Some academics are driven people and entirely focused on their research. Their personal lives may suffer for that and their students may suffer too. This kind of advisor may expect you to be like him/her in that way as well. The advisor may expect you to be on call all the time, may not be supportive of your outside interests or even of your family obligations. You can generally find out if an advisor is like that from other students. If you are really driven and completely focused on your research, you may thrive with such an advisor. Most people are not so driven; be warned [14].

  If you've done well in course work and your preliminary exams, advisors may come to you and ask you to work with them. That's flattering and may well be a good RA opportunity, but you are not obligated to accept a poor fit or an advisor who does not treat students the way you want to be treated. You are not a serf.

- Is the potential advisor near retirement? While many people are productive in their 60s, far fewer are in their 70s. You do not want someone who will retire or die while you are working on a project with them. It is fine to ask a potential advisor if she/he is planning to retire within five years. Many "mature" advisors (like the elder author of this book) stop taking new students a few years before retirement. However, some advisors seem to think that they are immortal and their students can suffer for that.

2.3 ▪ The Student's Job

We discussed some of your obligations to your advisor and your project in § 2.1. There is more to say about that and you have other obligations, some of which are to yourself.

2.3.1 ▪ Coursework and Qualifying Exams

Your first year in graduate school will be consumed with coursework. You must take that seriously. The grades you get will affect your chances of getting the advisor or project you want. Although this was briefly touched on in the introduction we will discuss these important components of graduate school here, too. During your first year and likely your second, you should be focusing on your courses. The department will assign a faculty member to assist you with course selection for that first year. Probably you will be taking core classes at first and then after you find a research advisor, there may be special topics courses.

Going to class, keeping up with the work, and doing well is expected. Grades in graduate school have a different interpretation. You should be getting A's. While a B or two is understandable (especially the first semester when there is a big adjustment), a C is not good and often *viewed* as an F. An F is unacceptable. A majority of students have to adjust the way they study when transitioning from undergraduate school to a PhD program. Know that it is normal to struggle a little while you figure out how to learn more independently. Also be aware that if your grades are not good, you may need to rethink if you are ready for a PhD program. Get help as soon as you think you need it.

You should be taking courses that prepare you for your qualifying exams. Take those exams seriously. Do not assume that because you got an A in a class, that you don't need to study. These exams are the time when you show you have what it takes to be an expert on a topic and that you are ready to be a PhD candidate. Having other graduate students to study and commiserate with can be a huge help. Always remember though that you need to be able to work out problems on your own. You should be doing as many practices exams as you can. Passing those exams will be one of the best feelings you'll experience on your pathway to a PhD .

2.3.2 ▪ Thesis Research

You get a PhD for significant and novel research. Your advisor's job is to guide you toward a project you can handle and help you along the way. Your advisor should be helping you with writing (see § 3.5.3), but will not write your thesis for you or solve every problem you encounter. Your degree is a certificate that you can do independent research and you will be doing most of the work.

Every thesis should have a thorough survey of the literature. Once you and your advisor have agreed on the topic, you must do the research necessary to demonstrate that you have mastered the literature at a level to both make your own contribution and convince others that your work is novel and significant. To this end, you should keep your supporting literature well-organized and build up your bibliography as you go.

You will almost certainly find that you need to learn new mathematics, a new computer language, or an application area. It is your job to do those things either through taking courses or independent research.

Getting started writing your thesis can be intimidating. Get a template from your department, read examples from other students, and plan to write frequently. Your thesis, although a technical document, must be in clear and correct English. If technical writing is not your strong point, all the more reason to start early and get feedback. A great resource is [26]. There are other good guides to technical writing aimed at general scientific communication [22, 23] and other fields [54].

There is nothing more stressful than waiting until the last minute to write up a project you have been working on for years. You should begin the thesis-writing process as soon as you have decided on a topic with your advisor. You may think that at the beginning that you have nothing to write about. You should get started by writing the background section which should include (but is not limited to) a review of the current state of the area. For example, if your thesis is about a new optimization algorithm you developed for a specific class of problems, describe the class of problems–why is it challenging, what other methods are traditionally applied and why is an improvement needed, what mathematical content is needed for a reader to understand the new work that you will be presenting later?

As you make progress on your research, you should be writing it up while the details are fresh in your mind. Your thesis might benefit from some well-designed figures. Before making your first figure, please read [50]. That book has many great examples of bad and silly figures and useful advice on making good ones. Schedule time to write and set up short term goals to get the content filled in. A good advisor will give you feedback on the quality of your writing and work with you to determine what belongs in the thesis. This is another reason to write and revise as you go. You can revise and improve your writing in small pieces. If you are well into your research and your advisor has not pushed you to begin writing, then you should speak up.

2.3.3 ▪ Teaching

If you have a teaching assistantship you must allocate the time to do your teaching well. You have an obligation to your department and, more importantly, to your students. Your teaching role can vary greatly from university to university but you will likely be in charge of teaching a class or some aspect of a class that is managed by a faculty member.

Effective communication with that professor is critical. For example, if that professor is the main lecturer three days per week and on the fourth day you run a class with the students, then you need to be aware of what was covered, use the same notation, and be in constant contact about what the expectations are. If you are a teaching assistant for a course you haven't had in many years, you should sit in on the lectures to get back up to speed.

You may be responsible for grading quizzes, exams, or homework. Do what the course leader expects and do it on time. If you have questions about an answer key or how to assign points, stop and get verification before grading 100 tests. That professor may have taught the class dozens of times and doesn't realize that you may need more support if you are new to teaching. Ask for that support if you need it.

When it comes to being a teaching assistant, here are some simple tips:

- The first day is the most important–you never get a second chance to make a first impression.
  - Introduce yourself and share your contact information and office hours.
  - Let them know your expectations (for example, no cell phones, being on time, etc...)
  - Get to know their names if you can (for example, have them put a folded piece of paper with their name on it on their desks)
  - Smile–foster a comfortable learning environment so they feel able to ask questions
  - Start *easy* with some review to build some confidence and not intimidate them right off the bat.
- Be on time.
- Be prepared–know the material you are going to teach and have notes.
- Talk to them–for example, if you are writing on the board, don't constantly talk to the board, turn and talk to them.
- Communicate issues with your professor–you will likely know the students better than they will.
- Do NOT slander the professor of the courses ever under any circumstances. The students should view you and the professor as a mutually supportive team who wants them to be successful.

For more tips see, [36].

If you are considering an academic career, you should try to teach even if you are supported by a research assistantship. It is wise to find a professor who can be a mentor to you in regards to your teaching. You will need a strong letter of recommendation when it comes time to apply for academic positions. If possible, try to teach a wide range of courses to build up your experiences. For example, having been a teaching assistant for Calculus II, Linear Algebra, and also Statistics is better than teaching Calculus for six semesters. If possible, try to get experience teaching a large class to make yourself even more marketable.

Particularly in the first year of graduate school, balancing your own coursework with your teaching can be a challenge. At your university, there should be guidelines about the teaching assistant work load. For example, a teaching assistant may be expected to work on average about 20 hours per week which includes their classroom time, office hours, course prep, and grading. If you are taking 20 hours each week simply to grade quizzes then you need to talk with the course professor about better time management strategies. Maybe you are leaving too much feedback?

On the other hand, the course leader is not allowed to exploit your services and expect you to work 40 hours a week for them. It is also not OK to put aside your teaching responsibilities for your research. If you find your research advisor is not respecting your responsibilities and your class is suffering, you need to say something. They may not even be aware. Do not hesitate to reach out for help if you feel overwhelmed.

Both teaching and coursework/research need to be priorities. Being successful in the classroom can feel rewarding during the times you feel like you are banging your head against the wall with your research and vice versa!

2.3.4 ▪ Professional Development

Graduate school is your opportunity to grow into your career. Your department may offer professional development seminars and you should attend those events. The topics are things such as giving effective talks, teaching well, writing papers, or dealing with deadlines.

Writing Papers

Students tend to have mixed feelings about writing papers ranging from terrified to excited. You will be partly responsible for writing the papers that come from your thesis. This is good news, because you are the expert. While your advisor may take the lead on this if she/he is a co-author, you should try to do as much as possible and learn the craft of paper writing from your advisor.

You will learn that a paper is very different from a thesis. Your thesis must include an exhaustive survey of the literature and detailed descriptions of work in the field. A paper, on the other hand, must respect the page limits of the journal and is typically limited to 20–25 pages. While your thesis draft may be a good beginning for papers, you will find that it's real work to modify thesis-style account of your work into a concise and clear publication for a journal. Fortunately, once you are well into writing your thesis you will have also read numerous journal papers. This practice reading papers positions you well to write papers.

Your advisor will (or should) explain the culture of your field, what journals are top-tier (such as SIAM journals), and how to select a journal for a particular paper. You'll need to know this if you are aiming for a research career and it is not easy to figure it out on your own. As an applied mathematician, you will likely be part on an interdisciplinary team. You may have the opportunity to publish papers in journals outside of mathematics. You will learn a lot during those sorts of collaborations and must remain mindful of the audience. Take the time to look through other papers, perhaps on related topics to yours, to get a better understanding of the scope and style of papers in a particular journal.

If you submit a paper while still in graduate school (and you should), your advisor should work with you to respond to the referees' reports. Most papers are not accepted without revision. The worst case is outright rejection. The typical best case is that the editor will send you two or more referees' reports on your paper and ask you to revise the paper.

In either case, you'll need to rely on your advisor's experience as you respond to the reports. A rejection may only be a signal that you've picked the wrong journal. In that case, fix any technical problems and submit your paper somewhere else. Referees' reports that ask for revision may sound hostile, and you advisor can tell you how to interpret them. You should never be upset by the tone of a referee's report. This is not personal, only business. Focus on solving the problems in your paper and not on your perception of the referee's attitude. Referees' reports generally make papers better and you should be grateful for them. Your advisor should remind you about this early and often.

It's very important that you have papers submitted before graduation and having some accepted or in print is better. If you want a posdoc, you should know that your competition has papers in print. You should also be formulating a five-year research plan as your work on your first papers. Your future employers may ask about that. Again, examples are out there to help guide you. Ask your advisor for support when you need it.

Collaboration

If you have the time, it is both good for your career and fun to collaborate on projects beyond your thesis research. You can get this started by attending conferences, talks in other departments, or simply hanging out with students from other departments. You'll find that many people like interdisciplinary collaboration.

If your collaborators have any integrity they will gladly list you as a co-author on any publications. This costs them very little and is a valuable boost for your career. Unfortunately, some collaborators have no integrity [5] and will use your work without giving you credit. You should negotiate this carefully before investing time and energy in a project outside your thesis work. The articles [16, 18, 5] have great stories about the rewards and risks of these collaborations.

Here is an infuriating story from [18].

> **Vignette:** *[18] "While I recognise that she thinks she had a major contribution to the project it will be difficult to show that she did more than a technician's job." This was the infuriating response I received in a long email thread with former colleagues about work I had done as an undergraduate researcher. During my 7-month internship, my colleagues in the lab told me I would be an author when they wrote up the work. A few months ago, I was shocked to learn that the resulting paper had already been accepted for publication—and that my contribution had been relegated to the acknowledgments section. This was my wake-up call about the need to speak up for myself regarding authorship, and to speak out against the unfair convention of diminishing the contributions of undergrads and technicians to scientific research.*

The idea that laboratory or computational work that contributes to a project is second rate is all too common. You must negotiate the rewards for your contributions outside of your own research group well in advance. Your advisor should be able to help you if your contributions are not recognized.

Giving Talks

If you plan to have a research career, you will need to give clear and convincing talks at conferences for your entire career. In any case, you will need to give compelling interview talks no matter what your career goals are. Getting practice is important. Your department may have a seminar for students to practice giving talks and you should try to participate. If your department does not have such a seminar, can you organize one yourself? You will find that the other students in the department will be happy to see you make the effort and will gladly participate. It's even better if you can convince some faculty mentors to attend.

A conference talk is different from an interview talk. We will discuss conference talks in this section and interview talks in § 2.3.6. Conference talks usually have tight time limits. At SIAM meetings the time limits are 20 minutes for the talk and 5 minutes for questions and setup time for the next speaker. It is rude to go overtime and puts the next speaker at a disadvantage. The chair of the session will, if he/she is doing the job properly, rigorously enforce time limits. Do not get on the session chair's nerves by running on too long. It is your job to find out the rules. They are usually easy to see in the call for papers at any conference. Your advisor can help you with this, too.

When giving *any* kind of talk the absolute most important thing is to know your audience. Applied mathematics is naturally interdisciplinary so you may present your work at a discipline-specific conference outside of mathematics. Depending on what your project is, the audience may consist of engineers, polymer scientists, or farmers and not necessarily mathematicians. Giving a conference talk is not about making yourself look like a genius by flashing complicated equations. Your goal is to present a clear message about the research results and why they are important. The important parts of giving talks include:

1. Stick to the time limit. Do not annoy your audience by talking too long.

2. Talk about one thing. Do not try to summarize everything you did in your thesis. Focus!

3. Here's a quote from John Dennis: *Never tell 'em more than they need to know.* Don't speculate about unrelated things in a talk. Don't tell people about things you did not or cannot do. These things lead your audience to places you don't want them to go.

4. Your viewgraphs should be sparse and in a font large enough for old people to read from the back of the room. The `beamer` LaTeXstyle enforces this and helps you get to the point without too many words or equations.

5. Do not have 45 slides for a 20 minute talk. The junior author has a personal rule that there should be fewer slides than the length of the talk (example, 20 minute talk means no more than 19 slides total).

6. The title slide needs to have more than the title and your name.

   (a) If your research has been funded, you should acknowledge the source(s) of your funding.

   (b) If the talk is about your thesis, you should mention your advisor. If you have co-authors, name them.

Rehearse your talk! Giving it to a mirror at home is not as good as having a live audience, but is far better than nothing. Rehearsals are one way to make sure you finish on time and that you say the important thing more that once. As you rehearse make sure that you are covering the main point (That's point, not pointS!) and you make it clear what your contribution is and how it fits in.

The short book [7] has many useful tips on presenting your results and the guidance in [26] is always useful.

Communication: Bottom Line

The bottom line for this entire section is this. Talks, papers, teaching, and your thesis are all part of developing good communication skills. If you speak and write clearly and directly, you have a significant advantage over the 95% of people who do not.

Email is one of the primary ways you will be communicating with your advisor and collaborators. Students often do not have experience professional email communication. Although it may seem trivial to even be including this here, it is easy to make a bad impression if you send an email that comes off as unprofessional. Keep the following in mind and note these also apply to in-person conversations:

- Be concise. Are your email messages long and verbose? Verbose responses to simple questions don't get read. An 80 line response to a yes-no question is not a response.
- Use correct grammar. Check that you used correct English in email. Do not use “textspeak" such as “u" instead of “you". Capitalize the first letter in a sentence. Use punctuation. This should all go without saying and yet the authors of this book still get emails written in textspeak.
- Include a salutation (“Hey" is not a good choice) and use people's appropriate titles (unless you have been told you can use first names). If you do not know if someone has a PhD and should be referred to as “doctor" or “professor" it is better to err on the side of caution to be respectful. It is sadly all too common that female professors are referred to as “Miss" as a default.
- Be careful about unintentionally hitting “reply all." (We have all done it at least once!) and be mindful about whether everyone on the email needs to be CC'ed. You don't want to clog up people's in-boxes.
- Write the body of your email before entering the recipients address in case you accidentally hit send before you are done.
- Use common courtesy such as “please" and “thank you".
- End with a closing

2.3.5 ▪ Funding Opportunities

Graduate Fellowships

Both the National Science Foundation (NSF) and Department of Energy (DOE) offer graduate fellowships. You, as the student, are responsible for most of the work for the applications. Your advisor will write a supporting letter and should help you prepare the proposal. You will need to explain your research plan, which is not an easy thing for a first or second year student to do, and your advisor's help is important. A DOE fellowship application must also explain how your project is connected to DOE programs.

The rules for these fellowships change annually. Your first step is to check the web pages
`https://www.nsfgrfp.org`
and
`https://science.osti.gov/wdts/scgf`
before you get to work.

Internships

Many national laboratories have internship programs. Sandia National Laboratories is one example
`https://www.sandia.gov/careers/students_postdocs/internships`

It helps if your advisor has connections at the laboratory and it is even better if someone at the laboratory is collaborating with you and your advisor on your thesis project. You should expect your job at the laboratory to be very different from your thesis project. An internship is a great opportunity to broaden your skill set, get serious parallel computing experience, and see if working at a national laboratory is a good career option for you.

You are representing more than just yourself while on an internship. You represent your advisor and your department. You are obligated to be professional. You must show up for work on time and be aware of your host's expectations. Poor performance can affect your advisor's future students' chances at an internship.

You should ask about help with travel expenses and housing. Most hosts have considerable infrastructure for that. Your office is likely to be a cubicle in a space with several other interns. This environment is a perfect place to build a professional network that will be with you for your entire career.

While the national laboratories always have summer internships, the availability of internships in industry has more variation. Internships in industry are a great experience and it's a good idea to look into that. SIAM has a useful web page about this
`https://www.siam.org/careers/internships`

Travel Grants

If you have an Research Assistantship (RA), your advisor may have some travel support for you in the grant that pays your salary. However, the budget for that is limited and you may need to agument that with other support. If you are on a Teaching Assistantship (TA) you may not be able to use your advisor's research budget for your own travel. The rules for this are not uniform, but you can, in general, only hope for travel support from a grant that pays your salary.

Be very careful about spending your own money for professional travel. § 2.4.2 has more details about the risks of doing that.

SIAM has a well-funded program for student travel awards `https://www.siam.org/conferences/travel-support/siam-student-travel-awards` If you are giving a talk or poster at a SIAM meeting, you should apply for support from SIAM. The application is simple and you need a short supporting statement from your advisor. We provide more information about SIAM conference opportunities in Chapter 4.

Many universities have travel support for students. These programs are often poorly publicized, which is to your advantage if you or your advisor know about them.

2.3.6 ▪ Finding a Job

While your advisor can and should help you with networking and your CV, you are the one who needs a job. You should have decided on a career objective long before you defend your thesis. If you did that, you're prepared to start applying. Most universities also have career fairs where industry and business representatives show up to look for employees. You should start going to these early on just to get a better understanding of what is out there and what employers are looking for.

You should prepare a professional web page and get an account on Linkedin (`https://www.linkedin.com`). Your web page and your Linkedin account page should be exclusively about your professional activities. Links to publications, presentations, home pages for any courses you've taught, and offices in professional societies are very appropriate for these places. Do not link to your personal social media accounts or include any personal information about yourself.

An interview talk is different from a talk at a conference. You need to tune the talk to the audience and, if you can, talk about how your skill set can be useful. It's a good idea to ask the person who invites you to interview about the audience and their level. Are you talking to managers, the technical staff, both? A talk to management does not need the level of detail that a talk to technical people does.

It is worth saying that your personal values should come in to play when thinking about your career (or choosing your graduate school for that matter). In applied mathematics, it is often impossible to predict where your work may eventually be used. Having said that, not everyone is comfortable working at a weapons lab while others are. In any job, management and culture do change over time and can become unexpectedly toxic. It is critical that you enjoy what you do. If you aren't a "people person" then probably teaching and advising students isn't for you. If you despise proving theorems and simply love coding, then there are jobs for you, too. When you do wind up with an on-site interview, be mindful of how it feels there. Ask yourself if you would be happy and follow your gut instinct.

Your CV

Your CV should be professional and tuned to the target organization. If you apply for jobs in industry, your teaching experience may not matter much. If you apply for jobs in four-year colleges, teaching experience matters more than anything else. The viewgraps at `https://www.siam.org/media/eova2f5e/2025-resume-webinar.pdf` link to a plethora of resources for developing your CV. The career center at your university is a place to start and you should also have your advisor give you feedback. Here are some general guidelines

- Do not include personal information such as hobbies and do not include a photo.
- Do not pad your CV–just the facts.
- Do not include job experiences that are not relevant (example–working at McDonalds in high school).
- Do include relevant course work, certifications, or microcredentials–tell them what you know.

- Do include submitted publications but do not include papers "in preparation"– those ideas can be included in your research statement.
- Industry/Lab CV should have programming skills (languages, operating systems, packages, . . .) at the top. Internships and publications are also important.
- Academic postdoc CV should highlight publications.
- Research university CV should stress publications and teaching experience. Internships also useful.
- Four-year teaching CV should emphasize teaching, working with undergraduates on projects, and any community outreach activities.

The Cover Letter

Can you write a business letter? Most likely you have not had to do this before but there is a first time for everything. The cover letters you write for job applications are business letters, i.e. formal communications between you and an organization. These letters are optimally short and focused. Selling yourself in a succinct letter isn't easy, but remember that your CV will likely be included with your application. You should not be cut and pasting actual mathematics or research results into the letter. A cover letter should include a header with the date, a salutation or formal greeting ("To the Committee" is good "Dear Sir/Madam" is not), the body/main content, a closing ("sincerely" is popular), and your signature and typed name. Within the body of the letter you need to state the position you are applying for and briefly highlight your relevant skills and experiences.

Use the job description to tailor what you include. What makes you a good fit? Avoid generic letters so that you can apply to jobs in bulk. You want to tailor each letter to the job you are applying for based on the job posting. Much like the CV, avoid sharing personal information. It is OK to mention that you have visited the place or grew up nearby but personal data is not OK. A professional tone is important. Never include negative comments about any previous employers or experiences.

It truly is important not to be too verbose. If you're filling up a page, you're going on too long by a factor of at least two. You should also know that verbose cover letters that fail to get to the point can be objects of ridicule for a hiring committee. That is not good for the candidate. Part of the cover letter's purpose is to make sure your application is filed in the correct place. A computer may be doing that filing. Make the filer's job easy by using only the words you need. If you are looking for a numerical analyst job, you do not want to wind up with a group hiring a existentialist.

It may be tempting to use a language model to write your cover letter. Do not do this. Your letter will look identical to that from all the other lazy people doing this. If your native language is not English and it is difficult to write a cover letter, there are resources help you express your ideas appropriately. Hiring committee members can tell when a letter is written by a computer. You will not make it past the first round.

2.4 ▪ If things go wrong

Up until now, we have discussed the roles and responsibilities of both the advisor and the advisee. Being aware of what the other one is dealing with and having ownership over *your* job is important. Certainly mutual respect and consistent communication are critical. Advisors should practice patience with their PhD students and embrace the fact that with the right mentoring, students will come into their own. Unfortunately, challenges and issues may arise, even when a good relationship is fostered and maintained. This section is about problems some graduate students may face that (1) are nobody's fault and (2) can be solved with some effort and attention. Chapter 6 is about problems that are the advisor's fault and are more serious and (we hope) less common than the ones we discuss in this section.

2.4.1 ▪ Dealing with Stress

You will have many responsibilities in graduate school. You must balance coursework, research, teaching, and your own personal obligations. It's important to think about this and figure out your own best approach to time management. For example, if you are on an RA, your advisor has every right to expect you to invest more time on research than if you have a TA. Your time management plan must reflect this. Another example is spending so much time on teaching that you neglect research. Most departments have people who can help you manage your teaching obligations. You should be talking with those people if you find teaching overwhelming.

If you decide that graduate school is not what you want or are having problems with depression or anxiety, you should consider leaving the university. If you think some time away would solve the problem, most graduate schools will let you take a leave of absence without having to fully withdraw from the program. You can then reenter the program without having to apply again.

Depression or anxiety can be very serious. Your counseling center is a very useful resource and there is no punishment for asking for help. Keep in mind that his problem is not rare, you are not at fault, nor are you alone [1, 15, 12].

Your advisor may be able to point you to resources, but most advisors are not qualified to help you deal with stress, anxiety, or depression. It is not fair to ask them to do that and what help they offer may not be useful. Your advisor is a mathematician, not a therapist. Advisors are however, human and should be respectful if you are dealing with any of these things.

2.4.2 ▪ Financial Problems

Your salary will be modest as a graduate student. You are investing your time in your education that will pay off later when you are on the job market. People enter graduate school at different times in their lives, although most transition directly after their undergraduate studies. It is usually not recommended to work a full-time job while you are in graduate school but some students are supporting a family or have reasons why they need additional income. You will need to be mindful of your own situation and finances. This is something personal that you will need to think about. If you budget wisely, and most graduate students do, you will be fine.

As you advance in your studies you may have the opportunity to attend conferences. Your expenses should be covered by university funds or your advisor's research grants. Some universities expect travelers to pay some of the expenses in advance and be reimbursed later. You should never lose that money, but may wind up with a larger than expected credit card balance.

When you travel ask your advisor or the person in your department who handles travel expenses if there are ways you can avoid committing your own money in advance. Many universities have resources for this.

The worst case, which does happen [45] is that the payment for your expenses is late. This is a particular risk if you will be reimbursed by the conference itself. There is, unfortunately, no solution for this problem. Neither your department nor your advisor is a bank and you should not ask them for loans. The conclusion of [45] is that you need to be aware of this risk and try to be prepared. You may have to decline travel opportunities if you cannot cover the up-front expenses.

Vignette: *[45] I can't access these funds until after the conference, long after paying for registration and flights. So, for the past 3 months, I have been out a significant chunk of cash. This is money that I can't really spare, as most of my modest graduate stipend goes to day-to-day necessities—including food, housing, and utilities. I'll submit my reimbursement forms as soon as possible once I get home. But I know from experience that sometimes payments are extremely delayed, even after the forms are filed. All I can do is hope the process will go smoothly this time—and know that I'll have to advocate for myself and micromanage every step if it doesn't.*

2.4.3 ▪ Problems with a project

You (or your advisor) may decide that the project or the collaboration is not working out. If you come to that decision early in the process, there is no loss if you find another advisor. If you wait until your fourth year or later, you could spend a much longer than usual time in graduate school. You will have to explain this to potential employers.

If you decide to leave your advisor, you should tell her/him that in person and explain why. Similarly, if your advisor thinks you not going to succeed in her/his program, the advisor should tell you face-to-face.

There are many things that can go wrong. These things need not be anybody's fault. You and your advisor should both be understanding when things like this happen.

- You may find that pressure, deadlines, and frequent reports are not for you. The sooner you leave a project like that, the better for you, your advisor, and the project.

- You may change your career objective and want a different advisor who can help you get to your new objective. Making a decision like this later than your second year can be extremely challenging so the sooner you come to this decision the better.

- You may decide that you simply don't like the project. Perhaps there is more or less computation/theory/analysis than you want. That's fine, but this is one more thing can you can't put off until year three.

- Your personalities may not mesh well. Both you and your advisor would be better off if you separated on good terms. This problem is the hardest one to figure out early, and it's possible that you'll be stuck if it's year four or later when you figure this out.

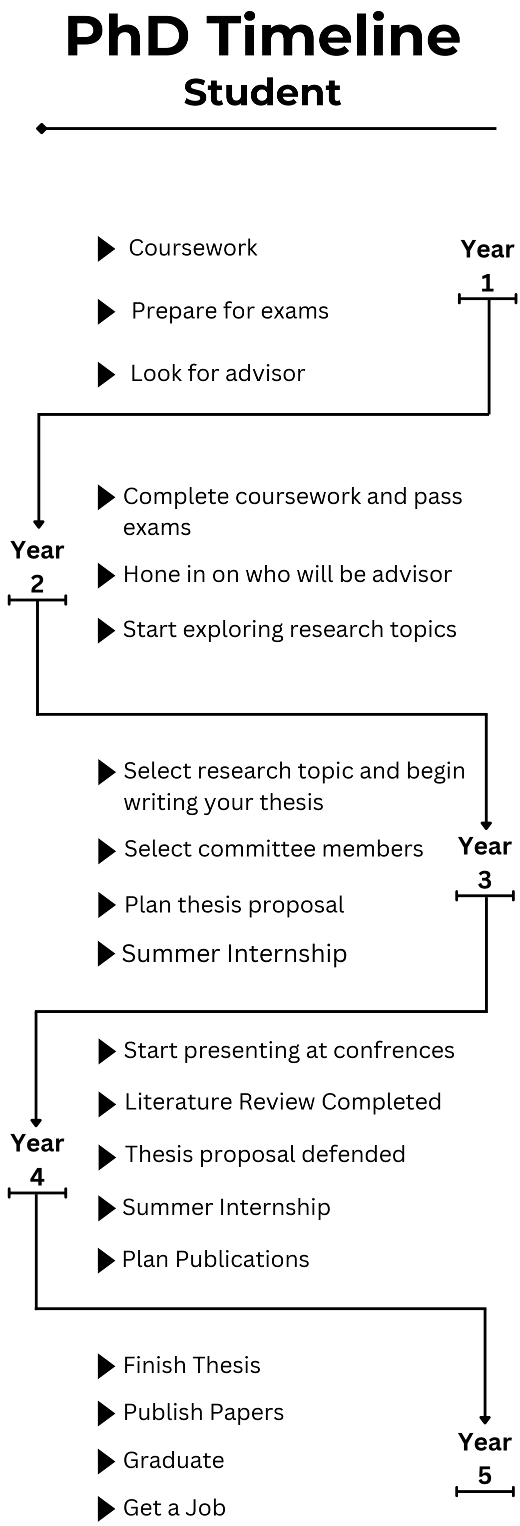
2.5 ▪ Final Thoughts

We emphasize throughout this chapter that student plays an active and central role in shaping the PhD experience. Success requires more than strong coursework performance: students must learn to manage their time, maintain professional integrity, and develop the independence expected of a future researcher. A large part of this involves building a strong, respectful working relationship with an advisor—one grounded in preparation, communication, and mutual expectations.

This chapter also discusses key decision points, such as choosing an advisor, navigating thesis research, and responding to inevitable challenges. Practical difficulties, from financial constraints to stalled projects, are discussed alongside vignettes that highlight the human side of graduate school. Importantly, students are encouraged to recognize when a project or advisor match is not working and to make timely adjustments to avoid prolonging the PhD unnecessarily.

Milestone awareness and pacing is key to success in graduate school! Figure 2.1 serves as a guidepost with expected progression through coursework, qualifying exams, advisor selection, internships, research, thesis proposal and defense, publication, and finally graduation and career placement. While each student's path may vary, consistent forward momentum is crucial. While the path through graduate school is demanding, the rewards are substantial. The persistence, discipline, and creativity developed during this journey are exactly what set PhD graduates apart. The ability to take a complex, open-ended problem, chart a path forward, and produce original, rigorous results is rare and highly valued both within and beyond academia. If the process were easy, everyone would pursue it—but it is precisely the challenge that makes the achievement meaningful.

Figure 2.1. *PhD Timeline*



Part II

For new faculty

Chapter 3
The Advisor

3.1 ▪ The Advisor's Rights and Obligations

It is of the utmost importance to treat your student with respect. Be mindful that they are engulfed in an intense learning process that can be stressful and challenging. It is your job to make sure they meet the important milestones while practicing patience and maintaining professionalism. To summarize;

- When students show up to meetings with you, be prepared. Hold them to the same standard and act quickly if they are repeatedly showing up unprepared.
- Be mindful of how much work you are giving your student. Take note if they are struggling with coursework, research, teaching and have authentic conversations with them about how they can adjust and balance all the demands of graduate school.
- Provide them with professional development opportunities as often as you can.
- Do not ask them to do personal favors, ever. They should not be mowing your lawn.
- Make sure that their efforts are leading them a thesis, a degree, publications, and a job. Provide them with guidance to do all of those things well.

3.2 ▪ The Advisor's Job

The student should be able to trust the advisor to do her/his job. This section explains what that job is. In summary, an advisor's job is to guide their graduate students through the PhD process which includes assisting them in selecting relevant coursework, providing some (but perhaps not too much) direction in research, reminding students of the timeline for finishing and reaching certain milestones, and mentoring them in regards to professionalism and job preparation.

As a junior faculty member you are competing with your senior colleagues for students. Although age is certainly not a guarantee about the quality of an advisor, more senior advisors have a track record that new faculty lack. Senior people with records of successful students are tough competition. Having said that, many new faculty members are full of energy and ideas and are eager to launch their research programs (usually with the help of some start-up funds!). Students figure this out and you should be (are will be expected to be) able to recruit and successfully mentor your students

As a new faculty member, being on the other side of student-advisor relationship for the first time can be exciting and overwhelming. It is critical to pay close attention to the policies and procedures at your new institution and to do everything possible to establish yourself as a successful advisor. Establishing a positive reputation will enable the recruitment of strong students and ultimately lead to a solid research program. Any advisor should be able to do those things well.

3.3 ▪ Thesis Projects and Recruiting Students

As an early career professor, one of your highest priorities should be to successfully launch your own research career. One of the first steps is to recruit students with high potential for success in research. You may have trouble attracting the very best students, who will be recruited by more senior faculty who can offer them RAs and have the contacts one needs to arrange internships. Having an exciting, emerging research focus along with a respectable publishing record and external funding definitely helps. However, being known as a sane, reasonable human being is also required. Students talk to each other and the word spreads rapidly about a faculty members' poor teaching, abusive behavior, or poor organizational skills. In other words, you want to work on establishing a *reputation* for doing quality work and for being approachable, organized, and reliable. We will discuss more about what that entails throughout this chapter.

3.3.1 ▪ Designing a Thesis Project

In order to recruit students, first you need to have a plans for possible thesis projects. This means knowing the scope of a PhD project. Although you will likely not have all the details planned out in advance, that is the nature of research, you should develop a loose timeline. Is it feasible to make significant progress in a few years? You should think about what skills are needed to advance the work, such as programming, an interdisciplinary background in a field of science, or statistical analysis tools. If a student doesn't have all the skills, you should have thought in advance about pathways for how they can get the right training such as taking special topics courses, attending workshops on parallel computing, or doing an internship. Remember, new graduate students do not know what the scope of a project *should* be like or what they are even getting into. This is your role as the advisor to have seriously thought about this in advance.

You will need to be well prepared when you talk to students about potential projects. You'll need to take topics from your own expertise and tune them to be at the level where a student can contribute in a significant way. This is not easy for a new faculty member. If your department has given you a senior mentor (and they should have), this is something to talk about. Good thesis projects are broad, a narrow technical point of limited interest may be a good topic of a paper for you to write yourself, but is unlikely to help a student land a job. The project should be easy to identify as applied mathematics. You should make sure that the results, if the project is successful, will be publishable and significant. You will also need to think about what to do if the student struggles with the project or it turns out that you've asked for the impossible. It is your job, not the student's, to make sure that if the student does what she/he is supposed to do, the project will result in PhD in a timely manner. Keeping a student in graduate school for six or more years because you failed to design the project well will (deservedly!) damage your reputation and make it hard to recruit more students.

When you talk to students you need to determine if they have the interest and ability to get up to speed on skills they do not have. For example, a student with a weak background in basic sciences may not want to invest the time in taking high level courses in physics, chemistry, or engineering and having to struggle with being unprepared. While it is usually the case that the student's background in mathematics can overcome many of these problems, the potential student may not believe this. You may need to direct such a student to a different advisor or find a project with different requirements. The same can be said about high-performance computing skills.

3.3.2 ▪ Recruiting Students

Once you have decided on an amazing research project that you would like find a student to work on, you have to be able to pitch it to a student. Develop an 'elevator talk'–could you describe the main idea in a few sentences? Think about how to promote your research and highlight how attractive and exciting it is. This initial communication about the project has to be packaged honestly and appropriately. Your goal is to help the student make an informed decision about something very important to their career pathway–what information does the student need to know to make up their mind? Take the time to plan out how you will describe the research, what will be needed to make progress, the timeline, and then what opportunities may open up as a result–such as publishing papers and developing marketable skills.

Let's suppose you have the idea and recruiting pitch in place–how do you get the news out that you are looking for a student? Teaching a graduate class is an ideal way to get to know your pool of students as well as establish the reputation we mentioned above. A new faculty member has the disadvantage that nobody in the student pool has any information on her/him. The best way to fix that is to teach courses for preliminary exams, and your department should give you that opportunity early in your career.

If a graduate class isn't in your teaching load when you are trying to find a student, ask another professor if you can go into their class a make the pitch. An effective way to really get the word out is to give a departmental colloquium or seminar talk, or even better–give a talk at a SIAM student chapter meeting. This allows for plenty of time and visuals so you can really give details about the work, lets the students see you in action and get to know you, and gives them an opportunity to ask questions in a comfortable, group setting.

Still, there are other ways to identify potential PhD students. As mentioned above, serving on the graduate committee in your department gives you an advantage for getting to see application packets. Alternatively, you can be in communication with the committee chair so they can be on the lookout for a potential good fit for your research program. Use your network of colleagues in academia–do some of your friends t other institutions have strong undergraduates applying to grad school? Although rare and depending on the size of your institution, it may be possible to recruit your own university's strong undergraduate students to stay and apply for the graduate program to work with you.

In general, a faculty member may have access to the student's application packet and, depending on how long the student has been around, they might be able to talk to some colleagues who had the student in class. However, it is a bit more of a gamble for the faculty member. A new professor may have to take a risk on a student who may have not performed at the top level in classes.

If you did a good job recruiting and pitching your research, you may have several students looking to work with you. Your department expects you to have PhD students, but that does not mean you have to accept all students who ask. You should see what other junior faculty do and avoid taking too many students at once. Most departments assign a mentor (such as a senior faculty member) to new professors. The quality of mentoring is not uniformly good. If your mentor takes that job seriously and does it well, that person will be useful as you learn to work with PhD students.

As you chose from a pool of potential students, ask questions like the ones we suggested for students in § 2.2. Here some other considerations.

- Did the potential student do well in course work? If not, are the courses where the student did not do well relevant to the research project? You do not need a student who struggles with the core mathematics one needs to work on your project.
- Does the student understand that PhD work is a job? Do they understand the departmental rules on length of time before preliminary oral exams, time-to-degree, and performance as a TA?
- If the student will be a TA, will she/he have the time to do the research you need them to do? Can you tailor a project that a TA can complete in a timely manner? That is a skill you can learn and your faculty mentor may help.
- If you have RA funding, does the funding agency expect frequent reports? Does the potential student understand that she/he will be contributing to those reports.

Ultimately, sometimes you will get a highly talented student interested in working with you and you can give them an in-depth project. Othertimes, you may *have* to take a student and you will need to think carefully about a project that they have the ability to finish. Know that if you do take on a student, you are primarily accountable for their progress and your management will expect you to graduate them on time. This book hopefully gives you guidance in doing that.

3.4 ▪ Guiding Research

Suppose you have pitched your project to a student who you vetted to be sure they were a good fit with you and they said yes. Congratulations! You have made a significant commitment in both time and energy and, as with any relationship, communication is key. One of the first things you will need to do is decide what the expectations are in terms of regular meetings and basic communication. It may be that they need to also attend meetings with your larger research group or start attending a special-topics seminar. You should take into account how much coursework the student has and if they are in the middle of studying for qualifying exams. Establishing a regular routine and then clearly laying out the expectations will help get the relationship started smoothly and provide structure for the student to adhere to. At the very beginning, it may help to brainstorm together about directions the project can go and also to outline a timeline of milestones; relevant coursework, special training (example, learning new software), conferences/workshops, the proposal, and final defense. Having a schedule will help the student commit to the work and stick to deliverables.

Upfront, you should lay out any ground rules you may have about communication. For example, you may want your student to check email frequently because that is your regular mode of communication. Just because you check email constantly doesn't mean that they do. You may want to exchange cell numbers so you can both communicate last minute meeting changes. However, if the student is reluctant to give you a mobile number, that tells you something about the students view of work-life balance. If you do not share that view, let the student know and encourage the student to find a different advisor. The senior author of this book thinks that it is absolutely improper to call students at their homes and has never asked for a mobile number and never given his away. Email is a far better venue for communication. There are no emergencies in academia that require calling someone at home. The junior author of this book regularly exchanges cell numbers with her students but is mindful and respectful about when to reach out in a text. In whatever form you choose, it is important for both the advisor and student to communicate expectations and boundaries up front to avoid conflicts. If you don't answer emails on nights or weekends, let them know. You should also be respectful of their needs. If your student is a parent, they should be able to tell you that they may be unable to respond to emails in the evenings.

How the weekly meetings are structured will evolve with the project but at least at the beginning, you should be explicit in what you want them to do each week. For example, consider the following;

- If you give them a paper to read, be specific with what you expect them to do. Should they come in with specific questions? Should they write a summary of what they've done? Should they be looking ahead to future directions?
- If they are implementing some code–what results/plots do you want them to show you? Be explicit with what you want.
- Students may be aware of technology that you are are not–be open to what they may bring to the table.
- If something goes wrong with the research–have the student quantify the problem. For example, maybe the results do not agree with the literature–by how much? in the 8th digit or by an order of magnitude?
- If results are late, get the student to commit to when they will have them done.
- Are you expecting them to write their work up on a weekly basis?

You may not even know this in advance but at the end of each meeting, you should be ready to advise on next steps, at least until the student has enough experience to make progress independently.

Most importantly, it is your role as the advisor to be prepared for these meetings and to give your student your full attention when they are there. Have them explain what their methodologies and findings. Listen to them and gauge how it is going. Be ready to answer questions (or say that you don't know–it is research after all). The point is, if you bumble your way through these meetings unprepared or unengaged, then you are wasting their time and yours. They have a finite amount of time to graduate. Remember, you made a commitment to advise them.

Vignette: *My first advisor always seemed distracted. He had significant administrative duties, and always seemed unprepared to advise me on our project. I tried every way I could think of to get him to understand what I needed help on. I would email him questions in advance of our meetings, I would make slides to show during our meeting to demonstrate my issues since verbal communication didn't seem to be enough. After some time, I realized that he just didn't have the bandwidth to advise me. It was difficult to figure out how to delicately switch advisors. I eventually had another faculty member willing to co-advise (with the understanding that he would just be advising and giving half credit to the other advisor).*

You do not want to be known as an abusive or negligent advisor. Here's a list of complaints that the authors of this book have heard about advisors. Don't let this happen to you.

- My advisor cannot (or will not) tell me when I can expect to finish.
- My advisor is unresponsive to my email, sometimes it takes a week or more before I hear anything.
- My advisor expects me to be on-call 24/7.
- My advisor shows up for our meetings unprepared.
- My advisor is completely dismissive of my original ideas and not willing to listen.
- My advisor seems happy with everything and doesn't push me to do more–there is no constructive criticism so I am not sure if what I am doing is actually any good.
- My advisor presents my work as their own and doesn't give me credit.

Of course, if your student is guilty of some of these things, you need to talk to the student and tell them that preparation and responsiveness are job-critical. Finally, we refer the reader to two other discussions of student supervision in applied mathematics [51, 48] which are very useful.

3.5 ▪ Writing and Defending the Thesis

3.5.1 ▪ The Proposal

We reiterate here that it is the responsibility of the advisor to know the university's policies in regards to completion requirements and to help the student stick to those. You *must do this right*. The student's career is at stake.

Timelines and expectations can vary widely. In particular the scope and the timing of the thesis *proposal* may not be what it was where you obtained your degree. You should attend as many thesis proposals and defenses as you can to get an understanding of the culture in your department. Better yet, get yourself onto someone's PhD committee. The proposal is done after a thorough literature search is in place and a research question is well formed and planned out, but the bulk of the research is not done yet.

At most universities, the proposal will consist of presentation of the research plan and progress to a committee chosen by the student and advisor and may or many not require a written component (thesis draft for example). Sometimes, it may be obvious who those committee members should be (or who to avoid completely!). Other times, you will need to think carefully about who the student should ask. There may be difficult trade-offs to consider in terms of balancing a person's expertise and their personality. Likely, expertise is more important. It may be that one committee member needs to be from outside the math department.

Again, the advisor should try to understand the culture of their department and the university to help with this important step. The student should be the one asking the faculty to serve on their committee. That request should include a short description of the research and a sentence or two about why that faculty member would add value to the committee. The student also, should be aware that it is a significant commitment for a faculty member to be on their committee, so they need to ask politely. You may want to see a draft of that email invitation first, but ultimately it should come from the student.

Regardless of what the nature of the proposal is like at your new institution, it cannot be emphasized enough that you, the advisor, need to do everything you can to support your student through this. It does not mean that you write their slides for them or give them solutions to oral exam questions. Those things are unethical. It means that you do not let them go through the proposal process until you feel confident that they are ready.

What exactly does *ready* mean? They need to convince their committee that they have what it takes to become an expert in a certain niche of mathematics. It means that they are able to explain how their work is new and adding value to the scientific community. They need to explain how it is different from pre-existing work. They need to have a timeline in mind about what work needs to be done for the project to be completed. It not only means that they understand the aim and scope research topic, it also means they can have an intelligent conversation about it. It can be devastating for a student to have a disastrous experience.

You should work with your student on selecting the advisory committee. That committee will have to approve the thesis proposal and the final defense and it's worth some effort to help your student pick a useful committee. If the project is a collaboration with another department, a national laboratory, or industry, then the collaborator must be on the committee. The student may have favorite teachers and it's fine to appoint these people if they are likely to be useful. A teacher known to be easy is not useful. You should also help the student avoid committee members who are needlessly aggressive. You want a committee that is tough and fair, not abusive.

Work with them on preparing for the proposal, giving feedback on their slides in advance and helping them anticipate difficult or probing questions. Also know that in the end, they need to be the ones answering questions and convincing the committee of their potential to be a researcher. At some point, you do need to step back and let them have ownership of their proposal and how the preliminary exam turns out. Let them know that it is ok to say, "I don't know" when they really do not know how to answer a question. An outcome of the proposal should be some feedback and suggestions from the committee about the project that will ultimately lead to improvement. If you have done your job right, the student will pass the preliminary oral exam. When a student fails this exam, and this is rare, *the advisor usually gets most of the blame*. If you let a student fail a preliminary oral exam, your career will (deservedly!) suffer.

Be sure to see if there is any necessary paperwork to complete after this milestone is reached. Now would also be a good time to review their coursework requirements to be sure they are on track with credit hours.

3.5.2 ▪ The Thesis: Getting Started

First things first, your student needs to find or be given your university's thesis template. It would especially help them to see a PhD thesis in math as they likely really have no idea what the expectations are. Again, seeing an example from your university is ideal. Although you have written a thesis, likely your student has not. Let them know that a clear take-away message from the thesis should be that the work is new and valuable. A thesis document will likely have the following content even though where it appears and how the chapters are organized may vary.

- An introduction that explains what the research is about, why is it important (motivation for the work), and how is the document organized.

- Some background about the mathematics and/or application since the reader may not be an expert in all aspects of the work. This part of the thesis should clearly and directly explain "who cares" and put the research in the broader context of science and engineering.
- Somewhere in the document there needs to be a literature review of related work and a clear message about how the research is significant and new. Citations are needed throughout.
- The "meat" of the project, which may include: explaining experimentation, simulations and software, analysis of data, new theorems/proofs, or numerical results. What was done?
- Strong conclusions (possibly summaries at the end of each chapter) that reinforce the relevance of the work. What is the take away message. You may need to say it several times in different places.
- Possible future directions.

One perspective is that it is really never too early to start writing. Most people need practice to become decent technical writers so the sooner anyone starts the better. Students may feel that they have nothing to write about at the beginning–not true. They can summarize the background literature and underlying methods the project is based on. Even if some of it gets taken out later, it is not time wasted since they will become better at technical writing. To save stress later, it is smart to have your student write *early and often*, with deliverables each week–even if it is just adding a page or two or formatting figures. There may be a learning curve using LaTeX. If a few weeks have gone by and the thesis document is not being updated or evolving, you should nudge them. It should be part of their work week to carve out a few hours to revisit that document and move it forward, even if sometimes that is with baby steps and other times it comes in the form of giant leaps. It needs to move forward.

3.5.3 ▪ Giving Feedback

If you are a skilled technical writer, it can be painful to read your student's thesis if they are not. This is not the fault of the student, rather just due to the fact that probably they haven't had an opportunity or reason to practice technical writing in mathematics. A great resource for them is [26]. If you get a student who is already a good technical writer–you are extremely lucky. Otherwise, it is your job to help them hone these skills.

You will need to learn how to give valuable feedback and be prepared to read many revisions. It is absolutely not OK to just re-write their work for them. It is easy to start marking up a thesis and replacing their sentences with your wording. Do not do that. They need to develop their own voice in communicating their research. Some suggestions are

- Point them to resources about technical writing with basic things they need to know (label axes, provide meaningful captions, number equations, etc...)
- Have them keep a nomenclature list so that, for example, h doesn't mean four different things depending on what chapter they are on.
- Use general pointers that lead them to improve their writing instead of fixing their work. For example, point out run on sentences, but you should not suggest exact revisions. Just write "grammar" in places that need fixing.

- If something that they wrote is just not clear, they may be confused or just not sure how to express an idea. You should go over those sections in your weekly meetings and get clarification. Simply writing something like "rewrite" or "not clear" over those parts is not helping them grow.
- Track changes if possible so that you know that they have addressed your concerns and how. You can begin to feel crazy suggesting the same edits over and over each time they give you back the document without the changes made.
- Be mindful of what sounds plagiarized, may need to be reworded, or needs a citation without accusing them of anything unless you see a pattern. Again, they may not know how to paraphrase or cite literature properly. If a student willfully commits plagiarism, tell the student (once only!) that this is serious academic misconduct. If it happens a second time, contact your department head and report the problem to your director of graduate programs. You do not want anything to do with a student like that.
- Do not hesitate to send them to your university's writing center if they need it. For foreign national students, writing a thesis in English can feel impossible.You can spend hours distracted by and trying to fix writing. This is not only problem for foreign students. Native speakers can also struggle with clear technical writing. There are professionals to help with that.
- However, if you send them to the writing center, carefully review the thesis to be sure the mathematical meaning was not lost in translation while a non-mathematician tried to fix grammar issues.
- Give positive feedback, too. It can feel bad (as a student) to get your thesis draft back with just markings all over it! Point out when something is well-written, too.

3.5.4 ▪ The Final Defense

At some point, the work is done and the final defense can happen. It is up to you and your student to decide exactly when this is, but all along you should have had a maximum of five years mapped out for the project and been ensuring progress was being made to reach that horizon for completion. Ask yourself, is the work publishable? Is it competitive with other work in the same area? Does the student have ownership over the content?

Be mindful of your university's deadlines for paperwork and graduation requirements which may influence when you need to schedule the defense. A committee should already be in place from the proposal but the talk will be significantly different. Many universities expect the committee to have the thesis well before the final exam. You should make sure your student meets that deadline.

Your student's slides should briefly recap some of the proposal ideas but focus primarily on what was done since the proposal. Similarly to the proposal presentation, you need to step back and let your student do the talking. This is their moment to shine.

3.6 ▪ Career Preparation

In addition to guiding research, the role of an advisor is to support their students as they become professional applied mathematicians. It is extremely rewarding to see your students head off into successful careers and for applied mathematicians, there is a broad range of possibilities. It can also feel like an immense responsibility. You want your student to be marketable and have the skillset to succeed without you. A strong research program will facilitate the numerous other professional development opportunities that the advisor should be nudging their students towards.

3.6.1 ▪ Writing Papers

Let's face it, if you are in a tenure track position, then publishing papers is at the forefront of your mind a majority of the time. For your students, not having publications on their CV can be a deal breaker as they are applying for jobs as well. The timing of publishing papers with graduate students can be very tricky. Sometimes, a project is a paper goldmine. In other cases, the main results may come near the end of the project, and the paper may at best be submitted by the time the final defense happens.

There may be opportunities along the way to publish intermediate results as peer-reviewed conference proceedings. These are often not valuable for tenure, but are low-hanging fruit and can help your student gain writing experience and improve their CV. Whatever form the publication is, you need to include your student's name if they contributed to the research or results at any level. If a paper is based on a students thesis, the student should be first author.

When you write papers with a student you should be generous with giving the student credit for her/his work. If the student did more than half of the work, we recommend that the student be first author on the paper. This is different from the standard practice in mathematics, but is common in other parts of science and engineering. You may do most of the writing, especially for a student's first paper, but it is still good practice to fully credit the student for the mathematics in the ordering of authors.

This is one thing where the student really needs your advice. In general SIAM journals are the best place to publish work in applied mathematics, but they have very high standards and may not be the right place for a first paper. If your work has impact in another field you should consider journals in that discipline. This is particularly true if you have a co-author from another field for whom publishing in the mathematics literature may not count for as much as it does for you. You want the paper to be read and picking an appropriate journal for the publication is important. The authors of a paper need to discuss this and the student needs to be part of that conversation.

You should explain the schedule for papers to your student. How many papers do you expect from the project? Will the papers be a steady stream or be submitted near the end of the project. Keep in mind that the students needs to have papers accepted while looking for a job. It is very important that you and the student communicate clearly on this. If the student expects to have papers in hand while looking for a job and you expect to submit the papers late in the project, then you must explain to the student why papers will come slowly and work with the student to get something out earlier. Clear communication on this can avoid bitter feelings at the end of the project.

Vignette: *I significantly wish I had changed advisors when I had the chance. My advisor was consistently disrespectful and borderline abusive. For these reasons, I wanted to change advisors. However, we had worked together for about a year, and it appeared we were close to finishing a publication. I was told we were probably two months out from finishing a paper. With the advice from the graduate program director, I stuck with the advisor, thinking that perhaps I would be able to finish up the publication before switching. To make a long story short, that paper was not submitted until my final year of graduate school. I later found out that this tactic was somewhat common with my advisor – publishing virtually everything right at the end of the PhD thesis. Then again, it might not have been intentional; my advisor might just be very bad at estimating paper completion time.*

When you get referees' reports, work with the student on the responses. This is an important part of the student's professional development. This is a good chance to teach the student that (most) referees are not hostile, even if a report looks hostile to the student, and that referee's reports require thoughtful and detailed responses.

Once the student graduates you should be very careful about collaborations. Papers with you after graduation will count less than other papers in evaluations. You should tell the student that it's time for her/him to establish an independent career. If the thesis project has unfinished work, it is ok to complete that work, but take care that a former student does not become an unpaid postdoc.

After several years, collaboration with a former student is no different from any other collaboration.

3.6.2 ▪ Networking and Conference Participation

Push your student to attend seminars and colloquia at your university and if possible, nearby universities. Part of your student's professional development should include participating in conferences as an opportunity for them to hear talks about cutting edge research, improve communication skills, get feedback on their research from peers and experts, foster their own network of collaborators, and gain a sense of what it is like to be part of the mathematical scientific community.

Vignette: *I love taking my students to conferences. I try to create a very "friendly chemistry" between us, and this leads to positive overall results. I believe that becoming my students co-traveler gives them a feeling of importance and self-confidence. This practice has resulted in better talks given by them, more interesting conversations (mathematical and otherwise) and I can say with certainty that (so far) only positive results have come out of it.*

The SIAM Annual Meeting and the meeting on Computational Science and Engineering have special events for students that include poster sessions, research tutorials, minisympoia, panels about careers, career fairs, and social events with peers as well as SIAM leadership. If your university has a student chapter, SIAM may provide funds to support this travel and there are also other student travel awards for conferences in general. You should be encouraging your students to apply for these. Conferences can be expensive–do not hesitate to ask your department chair what sort of support is available for student travel (you don't know if you don't ask)! Sectional meetings of the various professional societies are also an option and usually less expensive because they are local. On that note, you should be encouraging them to join professional societies, SIAM in particular.

Depending on your student's skill set (and confidence), they may want to start by delivering a poster–which is a little less intimidating and actually generates much conversation and potential for collaborations. They could also give a contributed talk to gain experience and eventually, if you are invited to talk in a minisymposium and your student is up to speed, you can ask the organizer if they can speak instead. You will certainly want to be sure the poster or talk is up to par before the student presents. If possible have them give a practice session to an audience or seminar at your university. This helps them gain confidence and will set your mind at ease that they are ready.

Conferences are just as much about networking as they are about hearing talks–so be an example to your students. If possible, introduce them to people in your field and include them in conversations. If you are comfortable and it is appropriate–invite them to a dinner with you and your collaborators However, you should not feel obligated to micromanage their schedule at the conference. We give more information about how a student can get the most from conference participation in Section 4.1.

Think back to your first conference and how it felt and what you experienced. Hopefully it was positive–but if it wasn't, what do you wish was different? As an advisor, you can influence the way your student views research and encourage them to become a leader in the mathematical community. Being active in conferences and including (encouraging?) them is an essential step in that direction.

> **Vignette:** *My advisor really encouraged all PhD students to go to conferences. I got the chance to join the first conference when I was still master student. Furthermore, my advisor motivated me to go to conferences and she gave some tips how to do the networking, when I asked her, because I'm really shy. Moreover, I wanted to do some further training for example to improve my presentations skills or to improve my didactic skills. So I asked her, whether I could join a certificate program from the university and she allowed me tp participate. This program was really helpful, I got new connections in our university and I believe it will help me with applications for new jobs. Furthermore, I was allowed to join a mentoring program, which also was very helpful to get in contact with other PhD students and learn form each other as well as in workshops with different contents, like e.g. how to make good career decisions, or how to do science communication.*

3.7 ▪ The Student's Job Search

Your student may or may not know what they want to do after graduation. Certainly internships, interdisciplinary projects, and research and teaching experiences can all influence that decision. You should be prepared to talk to them about pathways to careers in academia, business, industry, or government–all of which are viable careers with a PhD in mathematics. You should also start to have these conversations when you begin working with the student and let the student know that figuring out a career objective is an urgent matter.

3.7.1 ▪ Academic Positions

If your student wants to go into academia–they have choices about whether or not to be at a four year, two year, or research university. These job packages typically require a CV, a teaching statement, and possibly a research statement on top of a meaningful cover letter that specifically explains why they want to be at the university that they are applying to. They likely have never had to write something like that before–if yours are good examples, it would help to see those–or to find them good examples. At a minimum–you should read these over and give them valuable feedback. If you have never been on an academic hiring committee before, it might even help to ask someone else in your department to look them over and then volunteer to be on a hiring committee so that you understand how things work from the other side of the hiring table and can be a better advisor to your students.

If your student is funded through a teaching assistantship and you have any influence over their teaching assignments–try to ensure that they teach a wide range of courses that vary in enrollment, level, and type of class (theoretical, computational, lecture, PBL etc...) Be sure that someone (possibly you) observes them teaching and can make meaningful comments about how they relate to students and present information. You might want to encourage them to get an additional mentor who can work with them on being an effective teacher and will be able to write them a letter of recommendation that speaks to their abilities in front of a classroom. Encourage them to take part in professional development opportunities that focus on STEM education as long as it doesn't detract from them finishing their PhD . They can join the SIAM Activity Group on Applied Math Education for example.

If they want to be at a research university, it would help *them* to gain experience in mentoring students themselves and also get familiar with the grant writing process. If possible, they could assist with an REU on your campus, oversee an undergraduate student who can help make progress on some aspect of their work, or coach students for the COMAP International Mathematical Contest in Modeling (https://www.comap.com/undergraduate/contests/mcm/). If they can go into a job interview with some experience in guiding other students in research, this will boost them above many of the other people applying for an academic position. If this is not possible, when they craft their research statements for the job packages, they should include aspects of their future research agenda that would be ideal for PhD or undergraduate to work on.

It is not typical that PhD students write their own proposals before they graduate, but even something as simple as applying for SIAM Student Travel Awards to go to conferences shows they have the capacity to seek out funds. They should think about which funding agencies would be interested in their work and be ready for questions about grant opportunities at their job interview. If possible, let them sit in on some of your brainstorming sessions if you are writing a collaborative grant. It will help them to see how ideas are developed, a research plan is mapped out, and then translated into funding requests. If you are willing, sharing a successful (or at least highly rated) proposal with them as an example for when they start their career could be extremely valuable.

Finally, you should point them to the Mathematical Association of America's Project NExT (New Experiences in Teaching). In the MAA's words, "Project NExT is a professional development program for new or recent PhDs in the mathematical sciences. It addresses all aspects of an academic career: improving the teaching and learning of mathematics, engaging in research and scholarship, finding exciting and interesting service opportunities, and participating in professional activities. It also provides the participants with a network of peers and mentors as they assume these responsibilities." (See https://www.maa.org/programs-and-communities/professional-development/project-next) SIAM currently sponsors two Fellows each year.

You are not obligated to be a role model for your students. If you are consumed by your research and a student aspires to a teaching job, it is ok if you do your job and respect the student's choices.

> **Vignette:** *"One thing I learned from my experience as a PhD student is that a PhD advisor does not have to be a role model for the PhD student as both a researcher and an educator. My PhD advisor was an excellent role model and guide for me in his capacity as a researcher. He understood well how to motivate me and made our weekly meetings into low-pressure events.*
> *On the other hand, I did not find my PhD advisor to be a teacher/professor/educator role model for me. His desk was a sea of papers about 1 foot high. I admired him for being able to pull papers out of that mound of paper without much trouble, but his desk made me cringe during our first few meetings.*
> *I also found my PhD advisor's teaching style to be too formal. He only lectured, he asked very few questions to his students during class, and he did not seem to be able to anticipate which parts of the course would be more or less difficult for us."*

It is not good to belittle a student's career aspirations. Your job is to help the student, not clone yourself.

> **Vignette:** *My advisor was aware that I was interested in teaching-focused positions and that to reach that goal, I needed to actually teach more often than was typical for his students. He was verbally supportive of this. It still became very clear, though, that he saw the work I put into teaching (and in my last year, into applications for full-time positions) as a distraction from research, the most important thing. At the same time, after I had accepted a job, he used the fact that I had gotten an assistant professor position straight out of grad school as an example of how good and possible it is to stay in academia to one of my academic siblings who was interested in applying for industry summer internships. It felt like he both invalidated and exploited my work to further his view of what his students should be doing.*

You should certainly share your opinions on careers with your students, but you should not insist that they take your advice.

> **Vignette:** *The senior author of this book strongly and loudly tells his students to avoid the professor business. The junior author of this book wisely ignored that advice.*

Research/Teaching Statements

When your student applies for an academic position, they will likely need to submit a CV, cover letter, a request for three letters of recommendation, and then a set of concise statements about their research plan, teaching philosophy, and possibly a diversity statement. These are challenging to write (especially because they are each meant to be short!) yet are an important part of their application packet. The easiest is likely the research statement. This statement should highlight the main results of the work the student completed (with references) and then outline future plans. It is important that the future work sounds realistically achievable and that it can be done independently (i.e. that the student will be taking ownership and not relying on their advisor). In applied math, it can look attractive if the research statement can identify other disciplines that may benefit from the work and serve as potential new collaborations. This is also a good place to mention how the research can be the focus of a PhD project for a future advisee or accessible as a possible undergraduate project.

The teaching statement depends on the student's teaching experiences. As mentioned above, hopefully the student has taught to some extent if they were looking to go into academia, but in any case writing this for the for time takes effort. The purpose is to make a convincing argument that the applicant cares about student learning. It should convey the message that they put significant thought into what the students in their classes experience and how they are assessed. It should highlight how those experiences and assessments build transferable skills for the workforce as well as gets students excited about mathematics. Trite statements like "I know all my students' names" or "I use examples in class" don't mean much and are obvious things any professor should do–so adding details that make the teaching style unique are more impactful. Consider instead, "I learn my students' names by having them fill out a survey about themselves and then bring it to my office hours for a chat–this way they also know where my office is." Or something like, "I use homework problems that require students to collect temperatures using sensors so that they get a first hand experience in dealing with noisy data sets–something they would face in the real-world" Again, it is highly likely that the student, as a TA, has *not* implemented these ideas in the classroom–that is OK. They can discuss their vision/philosophy or how they would teach a certain topic.

3.7.2 ▪ Business, Industry, and Government

An application for a job in industry or government is not the same as as one for academics. In particular, different skills should be emphasized on the CV. Teaching experience is not of interest and should appear on a CV only in the employment history. On the other hand, computer skills like languages, operating systems, parallel computing should have a prominent place in the CV.

It's important to talk with students early about their career plans. If a student wants to work in industry of government, it makes very little sense to burden them with unnecessary teaching responsibilities. While a student may need a teaching assistantship for support, she/he does not need to participate in extra programs designed for those who aim for academic careers. This time is better spent doing things, such as taking courses in other disciplines and developing better computational skills. Publications are important and demonstrate that the student is productive.

The advisor's network is very important, much more so that in academics. If you had successful internships, the connections you made there can help your student. If you help your student find internships in government or industry, that is even better and your student has her/his own network to help.

As a new faculty member, you may be just starting to build your own professional network and get some visibility. The sooner you do this, the better for your students. There is no mystery in this. Attend conferences and get the community's attention. Do work that people outside of your academic circle care about. Publish some of your work in journals from other disciplines. Doing these things will get the visibility you need to help your students find internships and jobs in government and industry.

3.7.3 ▪ The Letter of Recommendation

You will probably be the person who knows your student the best in terms of their ability and potential. Your letter will be critical and will take time to craft. It needs to be an honest assessment of how the student will perform at the job they are applying to based on a description of what makes them qualified. You should be specific and highlight how the student's abilities address the needs of the position. Your letter should include a description of the following

- their main research contributions to the field,
- how they contributed to the research (for example, they took the lead on proving convergence criteria),
- special skills that make them right for the job (for example programming languages)
- how they played a role in a broader, interdisciplinary research team or at an internship (for example, they were able to gain expertise in hydrology in order to advance the modeling) ,
- how they behave professionally and how you observed this (for example, they gave a talk at a conference)
- how they performed in your course, especially any special topics courses or any special training they had and why it is important to the job,
- any service roles they took on in your department or university (for example, SIAM student chapter treasurer, AWM student chapter president,...) and any special outcomes from that (for example, they spearheaded a new seminar series in machine learning),
- evidence of effective teaching in a wide range of courses (if applying to an academic job),
- evidence of leadership, originality, independent thinking, creativity, innovation, motivation, perseverance, strong work ethic, teamwork, and exceptional communication skills.

On the other hand there are certain things that have **no** place in a support letter. Some examples are

- a long story about who YOU are and all of YOUR achievements,
- details of the mathematics underlying the research (for example, the proof of a theorem from their thesis cut and pasted from a paper–you should be able to summarize its relevance in your own words)
- evidence of implicit bias (for example, would you use the same phrasing if you were writing about a man versus a woman? "She has a lovely personality" has no place)

- lies–you need to be telling the truth about the student's abilities,
- detailed personal explanations of things (for example, "Jesse had a baby during her time here so it took longer to finish her thesis than expected." The student can reveal those things in their cover letter or interview if they see fit,
- discussion about the student's gender identity, race, or religion.

The bottom line is that you cannot dial it in on this letter. If you had a positive experience with your student, you will not have trouble saying great things about them. You will probably even enjoy reflecting on their growth and the success of the project. Hopefully that is the case. However, you cannot write a good letter for a bad student–your reputation is on the line. There may be unfortunate cases where you have a bad experience with your student and do not have positive things to say in a letter. Perhaps your student doesn't even ask you. If they do, the right thing to do would be to keep the letter as succinct as possible and only state the facts; you worked with the student for a specific time frame, state their thesis topic and any positive outcomes, express that they completed coursework,...and that may be it. One very good guide to writing effective letters is [32].

3.7.4 ▪ The Student's Job Talk and Interview

Hopefully your student gets a few interviews so that they have choices. For them (and as you may remember), there can be lots of anxiety associated with these interviews. You should be prepared throughout your advising career to help students hone their soft skills. As mentioned above, not everyone is good at giving presentations. It takes practice and being keenly aware of who the audience is. If your student has to give a job talk (and for academic positions–they definitely will), then you should plan to spend some time assisting. We discuss giving talks in more detail in § 2.3.4. Both you and the student would be well advised to read [31].

If the student is applying to an academic position, the presentation will need to be for an even broader audience than if it is for an industry/government position. The audience will be the entire department as well as graduate and possibly undergraduate students. Being able to communicate complicated ideas at a low level and captivating the audience is critical when applying to an academic job, since these are skills a professor should have. Moreover, instead of just focusing on key research results, the talk should include a few slides about potential directions for the research to go–since these would be projects that the students there could take on. This talk more than any other talk needs to be rehearsed in front of a live audience (try to get folks who are *not* already familiar with their work), given constructive criticism, and then rehearsed again.

Advise your student to really dig into the university's webpage where they are going and to be prepared to answer questions like

- Why do you want to work at this university?
- What sorts of funding opportunities to see for your research?
- Which journals publish this sort of research?
- How do you see your research interests evolving over the next five years?
- Who on our campus do you see yourself collaborating with?

- Which classes are you interested in teaching?
- Are there classes you are interested in developing?
- What was your best/worst teaching experience?

Moreover, you can help them come up with a list of questions that they should be asking such as

- What is a typical teaching load/class size?
- What is a typical start up package like?
- Is there support for attending conferences or sending students to conferences?
- What are the requirements for tenure–what is valued most?
- What are the expectations for bringing in funding/grants?
- What support is in place for new faculty in terms of writing grants?
- What are the students like here?
- How easy is it to make collaborations within the department and across other departments?
- What are the computing resources like?
- What is the timeline like for your decision? When will I hear back?

3.8 ▪ If things go wrong

Things can go wrong in the student-advisor relationship. Many of these are nobody's fault and the advisor should deal with them professionally. We talk about many of these things in § 2.4. While it is not your job to be a counselor, you should be prepared to direct students to your graduate program director or department head for help with stress, finances, and poor grades.

You must, however, take charge if there are problems with the thesis project. If you find that the student is failing to show up for meetings, shows up unprepared, or demonstrates lack of interest in other ways, you may want to tell the student to find another advisor. Saying something like "You don't seem to be interested in or committed to this project. It would be best if you found another advisor and a project you like better." is one professional way to handle this.

Vignette: *I was told that I should find a new advisor because he did not like me and did not want to work with me anymore.*

If a student is not happy with the project, the student may tell you that she/he wants to find another advisor. Your job is to wish the student the best. This may be hard if you have committed significant effort and funding to the student and your collaborators are depending on the student's finishing the project. Try to keep any anger and frustration to yourself, but do not feel terribly guilty if you fail.

Chapter 6 is about how students may encounter very destructive advisor behavior. The specific issues are bullying, fraud, racism and bias, and sexual harassment. If you are a decent human being, none of the warnings in that chapter should have anything to do with you. If you are guilty of these things you can lose your job and, in extreme cases, go to prison.

3.9 ▪ Final Thoughts

Being an advisor is far more than supervising research—it is about guiding students through a formative stage in their careers and lives. A strong advisor understands that their role includes mentoring, modeling professionalism, and helping students stay on track with the milestones of the PhD process. This means providing direction when needed, but also allowing space for students to develop independence and resilience. Striking that balance is one of the greatest challenges and responsibilities of advising.

Successfull advisors cultivate reputations not only for research excellence but also for fairness, reliability, and approachability. Students talk to one another, and word spreads quickly about faculty who create supportive environments versus those who do not. New faculty, in particular, must learn to establish themselves as both competent researchers and effective mentors. Doing so enables them to recruit talented students, launch productive research groups, and build sustainable careers.

Finally, advising is a reciprocal relationship. While the advisor sets the tone, both advisor and advisee share responsibility for maintaining clear communication, setting expectations, and navigating challenges. At its best, this relationship becomes a professional partnership where both parties grow—the student in research maturity and independence, and the advisor in leadership and mentorship skills. The true success of an advisor is measured not only in publications but in the careers of the students they help prepare for the next stage of their lives.

Just as the PhD timeline in Figure 2.1 outlines the student's progression from coursework to graduation, the advisor's role is to help keep that trajectory on track. Advisors who are mindful of this bigger picture can provide the right kind of guidance at the right time, ensuring that students move steadily toward scholarly achievement and professional success.

Part III

For everyone

Chapter 4
SIAM

The professional home for applied mathematicians, computational scientists, and data scientists in the United States is Society for Industrial and Applied Mathematics (SIAM). SIAM connects mathematicians, practitioners, scientists, engineers, educators, and students from across the world in ways that benefit everyone. At the time this book was written, SIAM is made up of an international community of 13,000+ individuals from over 100 countries. SIAM evolves to suit the needs of the scientific community by branching out in new research directions as they emerge. SIAM is well known to publish some of the most respected journals in topics covering the spectrum of applied mathematics as well as hosting a range of highly-attended, focused conferences.

To meet the broad, ever changing needs of applied mathematicians, SIAM has smaller, organized communities called SIAM Activity Groups. For example, areas include Data Science, Dynamical Systems, Optimization, Life Sciences, and Applied Math Education just to name a few. For the complete collection, see `https://www.siam.org/membership/activity-groups`. SIAM Activity Groups provide an intellectual forum for SIAM members to collaborate. Most activity groups organize a conference held every other year, organize special sessions at the SIAM Annual Meeting, and send out newsletters. Some even align with journals associated with their focus. Many offer prizes for contributions in the mathematical area. SIAM has an online platform that facilitates communication between the members of each activity group to announce events and share opportunities. Student membership includes free membership for two activity groups (typically this is a modest fee for regular members).

You should be a member of SIAM. Student memberships cost nothing and there are discounted rates for early career mathematicians. To join SIAM go to `https://www.siam.org/membership/join-siam/individual-members/student`. SIAM membership will help you in all aspects of your career. You may meet the person who will give you your first job at a SIAM conference. You can build the network of colleagues and friends that will be with you for your entire career by participating in SIAM. SIAM members get discounts on books, journals, and conferences. These discounts are the least of the benefits.

We have mentioned SIAM elsewhere in this book with links to their resources for professional development. Those links are included in this chapter too. The sections in this chapter discuss how to use and benefit from the student activities and professional development programs in SIAM and SIAM conferences.

4.1 ▪ Conferences

SIAM conferences are the optimal place for anyone in applied mathematics to present research and the environment at SIAM conferences is very welcoming. These conferences provide a venue for people to share their ideas and network to foster the growth of applied mathematics research. Most conferences are made up of plenary sessions, minisymposia of focused invited talks, general contributed sessions, and oftentimes a poster session. Larger conferences may have workshops, tutorials, social events, or special sessions to award prizes. Naturally all SIAM conferences have coffee breaks to facilitate opportunities to catch up with colleagues or make new connections. SIAM maintains a conference calendar that lists conferences that are scheduled over a year out. It is important to keep on top of deadlines for registration or participation. We explain in more detail some of the ways to participate below.

4.1.1 ▪ Presenting at SIAM Conferences

There are three standard ways to present your work at SIAM Conferences–contributed talks, minisymposium talks, and posters. You must submit an abstract to SIAM for a contributed talk or a poster (typical guidelines are about 1,500 characters). The format of a contributed talk is usually 15-minutes with an additional 5 minutes for discussion. SIAM rarely turns a proposal for a talk or poster away but acceptance is based on compatibility with the conference themes. When you submit an abstract for a contributed talk, the conference organizers will place you in a session with other speakers who have a similar focus under a broad theme. That session may or may not make complete sense for your talk, but the organizers do their best. Grouping hundreds of talks into sessions is not easy but this allows them to accept most contributed talks.

A minisymposium is organized by one or more people around a more specific theme and talks are usually 25 minutes with 5 minutes for questions. Minisymposia are collections of one to three sessions and the organizers invite the speakers personally. If you are invited to a minisymposium, accept the invitation. Minisymposia offer more visibility than either posters or contributed talks. Junior, but post PhD, researchers can improve their own visibility by organizing a minisymposium with three senior people and themselves as speakers. Do not be timid about inviting senior people to your minisymposium. Nobody gets upset by an invitation. Both authors of this book did this several times early in their careers.

SIAM conferences have poster sessions, typically in the evenings. These often include food and drink to encourage people to come. If you have never presented at a SIAM conference, a poster is a fine way to get started. SIAM's website has excellent resources to help you;

`https://www.siam.org/conferences-events/about-siam-conferences-and-events/conference-guidelines/guidelines-for-meeting-participants/`

Workshops, tutorials, and plenary talks are managed by the organizing committee. If you are asked by someone senior to help with a workshop or tutorial, do it. You will gain experience in communicating, networking, and having a professional presence. It also looks good on your CV.

4.1.2 ▪ Panels at SIAM Conferences

Many SIAM conferences include special sessions with a panel of experts who share their experiences in a certain topic and then answer questions from the audience. The

SIAM Industry Committee and the SIAM Career Opportunities Committee does a great job at organizing to help bridge the gap between academia and industry, communicate emerging research needs, and inform students about what they are looking for as employers of applied mathematicians. This information can also guide faculty members to develop or adapt courses that better align with workforce preparation. Junior faculty and students should attend the panels about industry or other selected topics that might be in the program. These panels may lead to opportunities for employment and collaboration. Even if you are in academics, collaboration with industry is a rich source of problems and reinforces your credibility as an applied mathematician. SIAM is the pace to make meaningful cross-disciplinary connections that can benefit your career.

The funding agencies in the United States send representatives to speak on panels at SIAM conferences. This is an opportunity for new faculty and also students to hear about what agencies are interested in funding. These panels give you a chance to see the program officers in person and ask relevant questions to improve your grant-writing skills. The program officers are happy to talk to you about their programs and steer you to programs that are a good fit for your research.

4.1.3 ▪ Networking

SIAM conferences offer invaluable opportunities for graduate students to connect with leaders and peers in the fields of applied mathematics, computational science, and data science. Networking at these events can significantly impact your academic and professional trajectory. Don't be shy – introduce yourself to faculty members from other universities whose research you admire, strike up conversations with professionals from industry about their work, and chat with fellow students about their projects. It helps to prepare a concise "elevator pitch" summarizing your research interests and career goals. Actively listen to others, ask thoughtful questions, and if it seems appropriate, exchange contact information to maintain connections after the conference. These interactions can lead to mentorship opportunities, collaborations, and even future job prospects. Take advantage of poster sessions, workshops, and social events to expand your network and learn from the diverse perspectives within the SIAM community. You should not be timid about approaching someone and asking them about opportunities. SIAM is a very open community and senior members are happy and excited to connect with and help you. This is particularly appropriate if they attend your talk or show up at your poster presentation. Conferences like these are not just about presenting your work; they're about building relationships that can shape your future in applied mathematics.

4.2 ▪ SIAM Benefits for Students

SIAM supports many students who are giving presentations with travel to conferences `https://www.siam.org/conferences/travel-support/siam-student-travel-awards`. Every student who plans to give a talk at a SIAM conference should apply for one of these. These are competitive, so they look great on your CV and of course the financial support is always appreciated. At the time this book was written, the awards come with free registration and financial support that is generous and substantial. Be aware of deadlines but the application process is very straightforward and will require a letter of support from your advisor.

In addition to free membership and the travel support mentioned above, students get substantial discounts on SIAM books. Two books of particular interest for students are

[33, 26]. The first book, [33], offers a deeper insight into several of the professional development ideas in this book (example CV, cover letters, and interviews) with a focus on how to seek employment in business, industry, or government agencies. The second book, [26], fully covers the publication process but is useful for anyone who needs to communicate mathematics. It teaches the fundamentals as well as tips for LaTeX and BibTex, version control, text editors, and tips for handling writing assignments that come up in jobs such as referee reports and book proposals.

SIAM sponsors student chapters `https://www.siam.org/students-education/student-chapters` at many universities. If there is not one at your university, it is easy `https://www.siam.org/students-education/student-chapters/start-a-chapter` to start one. At the time this book was written there are over 215 student chapters across the world

SIAM student chapters are a fantastic way for students in applied mathematics (and other related disciplines) to network with each other and design activities that meet their own needs. Some possible activities include organizing panels (or afternoon tea!) to learn more about faculty research, providing undergraduate majors with a panel about graduate school, practicing research talks or hosting an internal poster session, inviting a guest lecturer (more on this below), learning a new skill together as a group, or participating in a competition. Each chapter is unique in its programming and scope. Student chapters can apply for SIAM funds to support their events.

Activity in a student chapter, such as holding a leadership position, looks good on your CV. However, there is a danger of spending entirely too much time on activities other than your research. Pay attention to this and if your activity in a student chapter is hindering your academic progress, step down and let someone else take the lead.

SIAM supports travel of student chapter leadership to the SIAM Annual Meeting. The annual meeting includes a breakfast with the student chapter representatives and the SIAM leadership. If you're invited to this, you will find that the SIAM officers are very interested in you and your chapter. You are the future of SIAM and the leadership of the society knows this well. It is also an opportunity to hear what other student chapters are doing and share highlights of your group. There are specific sessions where student chapter representatives present their research. Many SIAM conferences have other special sessions for students, especially the annual meeting and the SIAM conference on Computational Science and Engineering. You can look through the most recent conference programs to see what has recently been offered.

SIAM frequently offers opportunities for students to engage in real-world problem solving through challenges, undergraduate research experiences, or graduate level week-long modeling workshops. We encourage you to explore the programs section of the SIAM webpage. At the time this book is written there are several programs that graduate students can apply for. One is the Graduate Student Mathematical Modeling Camp (GSMMC) and the other is Mathematical Problems in Industry (MPI) Workshop. Both offer valuable training and career development experiences for graduate students. The GSMMC is a four-day summer program focused on training in mathematical modeling and problem-solving skills relevant to industrial research. Participants work in teams under the guidance of faculty mentors to model and analyze problems. The MPI Workshop brings together researchers from academia and industry representatives with research and development projects requiring mathematical solutions. Workshop participants collaborate in groups comprised of faculty, scientists, postdocs, students, and industry representatives to address these problems. The week culminates in presentations and written reports–just the practice and experiences needed to be successful in graduate school and at your future job. These programs provide practical experience in applying mathematical techniques to real-world industrial problems but with less of a commitment compared to a full internship. They also offer valuable networking opportunities and potential pathways to internships and employment. There is some financial support to attend these so you should be checking the website for application details and deadlines. SIAM has links to other internship opportunities as well; `https://www.siam.org/careers/internships`

SIAM News is the bulletin of the society. SIAM News reports on SIAM elections, conferences, and publications. It also publishes technical articles about applied mathematics. These are aimed at the broad SIAM membership.

SIAM News is also a resource for graduate students. It offers a broad perspective on the field and its diverse applications. Emerging research, novel uses of mathematical methods, and real-world problems being tackled by mathematicians are highlighted and written so that all SIAM members can get a taste for what is happening in areas other than their own. *SIAM News* keeps the community up to date with the latest advancements and emerging trends in areas where mathematics is being applied–such as biology, medicine, physics, and data science. Beyond research, *SIAM News* also provides content relevant to career development, science policy, and mathematics education, offering insights into the professional landscape and potential career paths. The editors are happy to hear about new ideas for articles and they welcome submissions from student members. This is a unique opportunity to showcase student chapter activities, research projects, or perspectives on issues relevant to the applied mathematics community. Contributing a short article to *SIAM News* or *SIAM News Online* is a fantastic way to gain visibility within the SIAM network, share your work with a broad audience, and enhance your professional profile.

4.3 ▪ SIAM Career Resources

SIAM is the place to learn more about careers in industry and to get the experience you need in order to land your dream job. The SIAM Career fair is an opportunity for contact with representatives of the national laboratories and industry. airs are typically held either at SIAM Annual Meeting, the SIAM Conference on Computational Science and Engineering, or virtually. A key aspect of SIAM's career support comes through its conferences. While not all SIAM conferences have dedicated career fairs, they do provide excellent opportunities for networking with potential employers. As we mentioned above, poster sessions, conference social events, and specialized workshops often facilitate connections between students and professionals from various organizations. These interactions can be invaluable for exploring career options, learning about industry trends, and building relationships that may lead to future opportunities. SIAM encourages students to attend conferences and actively engage in networking to leverage these benefits for their career advancement.

A popular book published by SIAM in *the BIG (Business, Industry, Government) Jobs Guide* [33]. *The BIG Jobs Guide* is a practical handbook designed to help students and professionals in mathematics, statistics, and operations research navigate the job market in business, industry, and government (BIG). It addresses key questions job seekers face, such as identifying marketable skills, crafting effective resumes, finding internships, and exploring diverse career paths. The guide emphasizes the growing demand for professionals in these fields and offers advice on how to prepare for and secure BIG jobs, including insights from professionals in the field.

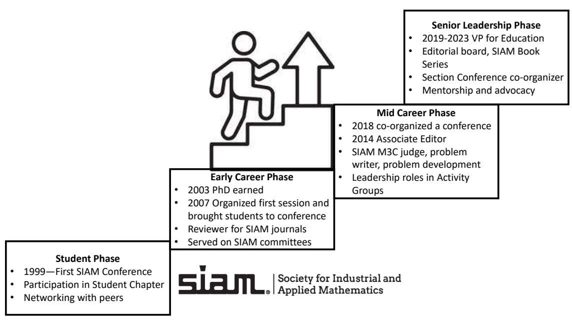
SIAM's website is constantly evolving with new resources about careers in applied mathematics including different career paths in applied mathematics, computational science, data science, and diverse opportunities available in academia, industry, and government. SIAM also hosts an online job board `https://jobs.siam.org` with opportunities in applied and computational mathematics. This is a much more focused list that others in mathematics and, therefore, more useful, especially if you aspire to work in industry or government.

4.4 ▪ SIAM Benefits for New Faculty

If you are a new faculty member reading this book, hopefully you already experienced some of the above mentioned benefits of SIAM as a graduate student. As a faculty member, it is critical to be part of the scientific community. All of the networking opportunities above are just as important, if not more important, when you are trying to establish yourself independently from your own advisor and convince your university that you deserve tenure. You are free to explore new areas that interest you and build your own circle of research collaborators. SIAM conferences are the perfect place to do this. Universities like to see their faculty collaborating with people outside their university. You may connect with someone at a SIAM conference who is a potential external evaluator for your tenure case. You may meet someone who invites you to give a colloquium at their university (add an invited talk on your CV!) You may connect with someone from industry and eventually apply for funding to work with them or send one of your own graduate students there for an internship.

Service (in addition to scholarship and teaching) is a standard part of the tenure evaluation. SIAM provides numerous opportunities for service activities. Refereeing for a SIAM journal is one service activity that also helps you to stay up to date on your research area. If you are good at being a referee and hand in quality reports, you may be asked to serve on the editorial board which looks fantastic for tenure.Simply organizing a session at a conference is another straightforward way to demonstrate your contributions to the scientific community. Finally, SIAM has a wide range of committees that need people to help shape the future of the society. Figure BLAH shows one of the author's SIAM trajectory as she progressed through her career. Involvement often begins as a student attending conferences, then expands into session organization, committee and editorial service, and ultimately leadership positions. Engagement with SIAM grows alongside one's academic and professional career, offering opportunities for networking, scholarship, and community building at every stage.

Figure 4.1. *Example of a professional trajectory in SIAM*



Chapter 5
Starting your Career

When you begin your career you will have to endure a process called *orientation*, *on-boarding*, or some other bureaucratic name. You may think that much of the time you'll spend in this process, usually several days, is a fine opportunity to catch up on sleep. It is not.

There are things you will need to pay attention to as you start your career that did not matter when you were a student. We will talk about a few of these things in this chapter.

5.1 ▪ Conflict of interest

Your new employer will have rules about conflict of interest. At a minimum you will have to let your employer know about most money you accept from another source. In academics it is not necessary to report payment for reviewing proposals or serving on panels for funding agencies **in the US** or honoraria and payment of expenses for talks or conferences **in the US**.

Recently activities outside of the US are coming under more scrutiny and you should be very careful to report **in advance** any income you will receive from institutions outside of the US. Your international collaborations are very likely to be fine, especially if you are in academics, but failure to report the income or the activity can get you in real trouble and even land you in prison [47].

Outside consulting for pay is generally fine with an academic employer, but you must report the activity and know your employer's rules about how much is permitted.

If you are working in industry or government, income from another source may be against the rules and you must understand those rules. Getting this wrong can cost you your job.

You may, as part of a federally funded project or part of your job, need to use *export-controlled* software. Pay attention to the training you'll get about this. You should never use export-controlled software in collaborations outside of the project without permission.

5.2 ▪ Workplace computers and technology

It is very likely that your new employer will give you one (or more) computers. Don't think of these as "your" machines—they belong to the company. These machines may have software installed to let the IT people examine your file system and monitor your internet traffic. Your employer may not allow any personal use of these machines or may understand that some personal use while at work is reasonable. In any case, do not put any sensitive personal information (health, taxes, financial, ...) on a work computer.

The same goes for email. Keep your work email for work and your personal email for personal matters. And as tempting as it might be, don't let your family use your work laptop—it's not worth the risk. If you need a computer for personal or family use, buy one yourself.

You will need to take training in things like internet security and other workplace policies. Topics usually include cybersecurity, internet safety, workplace policies, and professional conduct. This stuff is very dull. However, if you get anything wrong after the training you can get in real trouble. At one national laboratory in the United States, and we are not making this up, the on-line cybersecurity training used to feature a man in a horse's head `https://www.amazon.com/KingmysÂő-Latex-Horse-Head-Brown/dp/B016B144M6` Take it seriously anyhow.

Beyond the tech side, orientation usually covers practical things like how to submit travel expenses, the rules for promotions, and workplace expectations. If you're starting a teaching position, you'll probably get an introduction to classroom technology. Expect to see course management platforms like Canvas, Blackboard, or Moodle. These systems handle everything from posting assignments to grading, and you'll be expected to know your way around them quickly. You may also need to learn how to use classroom presentation tools, clicker systems for quizzes, or even online proctoring platforms if your department offers hybrid or online courses. This can feel very different from the tools you were used to in graduate school, where you may have just uploaded a PDF or handed out paper homework.

It can be overwhelming at first, but don't ignore these resources—they're part of how students experience your teaching. A well-organized online course page or a smart use of classroom tech can make you look polished and professional (and save you time in the long run). Many employers also assign a mentor to help you settle in. If you're lucky enough to have one, ask them to show you the tricks they've learned about managing classes and using the available technology. A good mentor can flatten the learning curve, helping you spend less time troubleshooting software and more time focusing on your teaching and research.

5.3 ▪ Your life savings

You will probably start your first job before you are 30 years old. The sessions on retirement will be the dullest and most soporific in the entire orientation event. The senior author of this book felt that way, too. He was wrong, but had some paternal guidance to straighten him out.

Please stay awake for this section. You will probably have a retirement package as part of your job and your new employer will contribute to a retirement account for you. These funds will compound **tax-free** for your entire working life. You may also have the option of contributing more from your salary before taxes. If you can afford to do this, **DO IT**! You may not be able to afford this early in your career, but as you advance and your salary grows, put as much as you can in the tax-advantaged retirement options from your employer.

Homework: Suppose you and your employer contribute $3,000/month to your retirement savings for a 40 year career. How much will you have at the end if the funds grow at a 5% annual rate? Use monthly compounding.

You may have the option of picking the investments in your retirement account. This may tempt you to try to predict the markets or, even worse, to put your retirement funds entirely in bonds to avoid the volatility in the markets. **DO NOT DO THESE THINGS!**

The senior author will tell you what his father told him. Listen to the old man.

> **Vignette:** *The words of Pa Kelley: Son, put your money in a broadly diversified mutual fund. Come back in 20 years and see how you did.*

- The "low-risk" strategy of putting your savings entirely in bonds is, in fact, the high-risk approach because you are guaranteed to get beaten by inflation.

- Do not try to pick "hot" stocks or time the market. Doing this is a fast track to poverty.

- Do not look at the value of your retirement portfolio very often. This is a long-term investment and monthly (or even annual) fluctuations in value are not something to worry about.

- Put as much as you can and are comfortable with in stocks. Index funds and broadly diversified mutual funds work well and you may have the option to do that in your retirement plan. 70% stocks and 30% bonds will be fine. Target date mutual funds might be better for you.

- Many retirement plans offer "target date" mutual funds. These investments put more of the account in bonds as you approach retirement. You put your money in the fund aimed at the year you plan to retire. The senior author of this book used a date five years later than he planned to retire because he wanted to be more invested in stocks.

- Do not churn your retirement portfolio. Do not buy or sell on market highs or lows. Do not panic when the market goes down, as it will. Do not get excited when it goes up.

- It is worth your while to learn something about investing. This is your life savings. It is important.

- None of the advice you get from books (including this one), a broker, the TV news, and the internet is the absolute truth. It is your job to filter this advice and figure out what is best for you.

Once you retire you must take a certain fraction of your savings annually. This fraction grows as you age. In your 70s it will be roughly 4%.

Homework: What is 4% of the amount you got from the homework earlier in this section? Compare that to the estimate of your social security income from

`https://www.ssa.gov`

5.4 ▪ Final Thoughts

The first steps of your career after graduate school can feel both exciting and overwhelming. Orientation, workplace policies, and new technology may seem tedious at first, but they're actually signals that you've moved into a professional world with new responsibilities. From conflict of interest rules to computer use policies, your employer expects you to operate with integrity and maturity. Paying attention to these details early prevents problems later and sets the tone for a smooth transition.

At the same time, this is when you begin to build your professional identity outside of graduate school. Whether you are in academia, industry, or government, you'll be learning how to navigate new systems, manage your own resources, and establish habits that will carry through the rest of your career. The effort you put into understanding the culture, using the tools provided, and taking advantage of mentorship opportunities will pay off in confidence and independence. Starting your career is less about knowing everything right away and more about being open to learning, asking good questions, and steadily growing into your new role. You got this!

Chapter 6

Bad Things

In this chapter we cover some serious issues that can be traumatizing to go through and could destroy a student's career. These are bullying, fraud, racism and bias, and sexual harassment. We do not intend this section to discourage you from a career in applied mathematics. Most students have fun in graduate school and go on to successful lives. You are likely to do the same. The Applied Mathematics Community needs all the good people it can find. Please do not become discouraged and leave the field because of the bad behavior of a few predators/bigots/bullies/frauds. These people are far outnumbered by the decent human beings in this community.

However, at some point in your career, you are almost certain to encounter people who are part of the problems we cover in this section. Your best bet as a student is to identify them and avoid them. Your fellow students can help. If you run into one of these people later in your career, please do the community a favor and help to get rid of them. If you are a victim of any of these things, do not remain silent. Your dean's office should be your first stop. Your dean is trained to deal with these problems and protect you.

Science is the flagship journal of the American Association for the Advancement of Science. Science reports all too frequently about problems such as bullying, fraud, and sexual harassment. Not too long ago, universities would try to cover these problems up and, in extreme cases, fire the offending faculty member without reporting to the larger community. The faculty member, especially if she/he was well known and well funded, usually had little trouble finding a new job. This is finally beginning to change.

For example, the National Science Foundation (NSF) has a policy on sexual harassment

`https://www.nsf.gov/od/odi/harassment.jsp`

Any organization must notify the NSF when an employee has been found to have committed sexual harassment and/or has been placed on administrative leave for sexual harassment.

The National Institutes of Health (NIH) also has such a policy [29] and has recently made the reporting requirements more rigorous [30], but not ass rigorous as those of the NSF. NIH has work to do on this and, as recently as 2023, has done poorly [52].

NSF and all US funding agencies also respond very forcefully to fraud.

`https://www.nsf.gov/oig/regulations/`

Academic misconduct can have very serious consequences. If the offense is serious and flagrant enough, people can be fired and banned from campus on very short notice. This, of course, leaves their students hanging.

> **Vignette:** *…their advisor got banned from campus. It was terrible. The university wouldn't tell the grad students why, and the chair was like, "Why should you care why?" Bizarrely, this is not a unique experience for grad students.*

If this happens to you, you should expect your department to help you get back on track. But, as you see in the vignette, there are no guarantees.

6.1 ▪ Bullying

Bullying is more, much more, than your advisor raising her/his voice when you don't do what you need to do in your research. You should expect your advisor to be annoyed when you fail to meet your obligations.

The online Science article [34] has a very clear description.

> **Vignette:** *[34] I think people can confuse academic bullying with academic freedom. Academic freedom means that you can pursue any kind of research you want and that you have the right to criticize any hypothesis in a polite and scientific manner. Often, however, people use this concept as a back door to justify inappropriate behaviors.*
> *Academic bullying happens because the academic hierarchy creates a large difference in power between principal investigators who fund and drive the research and the lab members who carry out the research. Bullying can consist of inappropriately changing authorship positions, taking credit for ideas or intellectual property for one's own benefit, verbally abusing others in a regular and progressive way, ridiculing or publicly shaming people, and threatening a bad recommendation or the loss of a job position.*

The damage done by bullying is serious and can drive people from their research. The author of [43] suffered because of her advisor's behavior and was driven from a research career. Her conclusion is strong and very sad.

> **Vignette:** *[43] I have struggled with how to think about this experience. I've found a fulfilling life in science communication and education, but I sometimes feel disappointed that I had to abandon my doctoral studies. I remind myself, though, that resigning doesn't mean that I "lost." I would have lost if I had accepted bullying as a rite in academe and sacrificed my mental health and quality of life. I know that taking on this bully was right for me, and I hope that speaking about my experience will encourage others to stand up for themselves, too. Doing so may put their career plans on hold or even at risk, but it's the only way to make academic life more egalitarian.*

In extreme cases professors can lose their jobs, but constant bullying, at any level, is toxic and you do not have to accept it. You can and should find another advisor and let your department head know about the problem. The sooner you get a bully out of your life the better. The article [41] is an ultimately positive account of how one student changed advisors because of bullying.

> **Vignette:** *[41] As I looked for a new adviser, I was careful not to repeat the mistakes I had made in the past. I approached another faculty member whose research interested me. But when I visited his lab, I paid attention to how he interacted with others, and I spoke with people who worked with him, specifically asking about his mentoring style. I was amazed how friendly everyone was. His students had great regard for him, and I immediately sensed that his lab was an environment where I could thrive. He took me on as a Ph.D. student, rescuing me from the misery I'd felt in graduate school up to that point. In my new lab, I arrived eager to work each day, and I didn't go home mentally exhausted by interpersonal disputes. That helped me be much more determined and efficient with my work.*

There has been progress. The article [17] reports on several cases where the offending professor has been fired or resigned under pressure. There has also been a recent case [39] in applied mathematics. Moreover, these cases are getting international publicity rather than being covered up. However, [17, 20] there is important work left to do:

Neither the US National Science Foundation nor the US National Institutes of Health specifically mentions bullying in its anti-harassment policies.

The Academic Parity Movement (see `https://paritymovement.org`) is an excellent resource for victims of academic bullying.

6.2 ▪ Fraud

As data become more of a part of applied mathematics, so does the opportunity for fraud. Students in other disciplines have been caught up in fraudulent acts of their advisors. Research misconduct is taken very seriously. Careers end and people go to prison for this. If you find that your advisor is committing fraud and do not report it, you are risking your career. You have very little choice. You must find another advisor and report the misconduct.

While only one [38] of the references cited in this section is about applied mathematics, the overall issues are independent of scientific discipline. The case reported in [38] is one of mishandling US federal funds. This is very serious and the professor's students will certainly pay a price for her offenses. At a minimum each of them will need a new project and a new advisor. Here is a a paragraph that illustrates how serious this is *...future is in jeopardy. On 21 May, she is scheduled to be in a federal court in Atlanta learning her punishment for mishandling a National Science Foundation (NSF) grant that supported her research. The judge's sentence, which follows her guilty plea in December 2019 to charges of providing false information to NSF, is likely to trigger a review by Georgia Tech officials that could lead to her dismissal.*

One point that you, as a student, need to keep in mind is that you have responsibility for accurately reporting your expenses. This includes travel to conferences, for example. Quoting [38] again *But her case is a sobering reminder that, although institutions are responsible for ensuring that grants are properly managed, individual scientists can face serious consequences if they don't follow the rules.*

Worse yet, you may base your thesis work on prior fraudulent work of your advisor. Even if you have no knowledge of the misconduct, your thesis results lose validity and the damage to your career will be significant. This New York Times article [3] is about such a case. Two students reported fraud by a senior professor. A university must investigate such reports and the findings were that the professor had been committing fraud for years. The part most relevant to you as a student is *At the end of November, the universities unveiled their final report at a joint news conference: ... had committed fraud in at least 55 of his papers, as well as in 10 PhD dissertations written by his students. The students were not culpable, even though their work was now tarnished.* The professor lost his job and may be under criminal investigation *Late last year, the Dutch government said it was investigating whether ... misused public funds in the form of research grants.*

Misuse of funding is almost impossible for a student to know about. The impact on your work, however, could be large. If your advisor is arrested or indicted, which has happened [47, 38], you may have to start over in your research.

On the other hand, no student should be committing fraud. This article [35] is about how fraudulent papers by a junior researcher damaged a large research enterprise. The onus was on the leader of the research group to clean it up, which he did. You have a responsibility to conduct your research ethically. The lead paragraph in the article puts fear into the heart of anyone who mentors junior people.

Vignette: *[35]* It is every scientist's nightmare: An editor calls saying he needs to talk urgently about a paper you've submitted. A reviewer thinks there's something weird in the data. It can't be too bad, you tell yourself, because you've read the manuscript a dozen times, and the smart people in your lab have been plowing this ground for 2 years. But when you start checking the reviewer's questions, you get a queasy felling. You dig through the freezer, bring out clones, repeat experiments. Soon there's no denying it: The evidence says you have been the unwitting co-author of a fraud. Now what do you do?

Finally, your advisor is responsible for reporting all sources of support to your university and to the government. The laws are becoming tougher on this issue [37]. If you receive support from a foreign government, for example, you must make sure your advisor and your department know the details.

6.3 ▪ Racism and Bias

If you are not a white heterosexual male, you will almost certainly encounter racism and/or bias in your career. You may feel pressure to act, dress, or speak in a way that does not reflect who you are [49] or feel obligated to listen to jokes that target your background, gender, nationality, or any of several other things [13, 2]. This kind of humor does not belong in a scientific environment (or anywhere else). Perhaps you will find that talking to the people making such jokes will solve the problem. Most people are well-intentioned, but perhaps insensitive. However, the barrier to speaking to the offender may be high, in which case you should talk to someone in your dean's office.

You may also find faculty members who relentlessly ignore you in class, belittle your ideas, or constantly say things that make you feel unwelcome. If other students have had similar experiences, you owe it to yourself and to others to make it stop. You do not have to tolerate this in silence and have every right to speak up. The atmosphere in the US has changed for the better on this issue and you will find more support than you would have even a few years ago. Having said that, you already know that the situation is far from ideal. **Katie, is "changed for the better" still true?**

Please speak up if you encounter racism or bias. Often a word to a professor or fellow student will solve the problem. Your dean's office is the place to start for more serious issues.

6.4 ▪ Sexual Harassment

If a professor asks you out, that is a violation of university rules. There is very little room to debate this. Telling the idiot to go away may solve the problem for you, but not for others. Reporting this to your department head or dean is one step toward protecting others. The impact of being a victim can be devastating and long-lasting. The article [4] reports that an undergraduate mathematics student and many others went through traumatic experiences that went on for years before the mathematician was dealt with. If a professor goes beyond that and pressures you for sexual favors, you have to act. The stories from [21, 57] show that, finally, students who take action on this can get the offender removed from student interaction.

According to the United States Equal Employment Opportunity Commission (U.S. EEOC), "It is unlawful to harass a person (an applicant or employee) because of that person's sex. Harassment can include sexual harassment/ or unwelcome sexual advances, requests for sexual favors, and other verbal or physical harassment of a sexual nature. Harassment does not have to be of a sexual nature, however, and can include offensive remarks about a person's sex. For example, it is illegal to harass a woman by making offensive comments about women in general. Both victim and the harasser can be either a woman or a man, and the victim and harasser can be the same sex. Although the law doesn't prohibit simple teasing, offhand comments, or isolated incidents that are not very serious, harassment is illegal when it is so frequent or severe that it creates a hostile or offensive work environment or when it results in an adverse employment decision (such as the victim being fired or demoted). The harasser can be the victim's supervisor, a supervisor in another area, a co-worker, or someone who is not an employee of the employer, such as a client or customer."

Other forms of sexual harassment are more subtle and others are more extreme than just being asked out or solicited. Closely relate to sexual harassment is sexual discrimination in which someone is treated unfavorably because of that person's sex, including the person's sexual orientation, gender identity, or pregnancy. In higher education, this is unacceptable and there is a law prohibiting this referred to as Title IX of the Education Amendments of 1972, with further amendments made in 2020. Every member of an academic institution must comply and there are consequences if they don't. Moreover, most universities require annual training for all campus community members so faculty should be aware of the consequences. Although not directly math-focused, the documentary Picture a Scientist (https://www.pictureascientist.com/) provides a grim overview of sexual harassment and discrimination in STEM and is adequately described as "quietly devastating."

It cannot be stressed enough that if you feel that you are a victim of sexual harassment or discrimination, that you say something. This can be easier said than done when you have been through something traumatic or feel threatened, but reaching out for trusted support is essential to the healing process. You can and should report this. If you do not, it will happen again to someone else. Karen Kelsky, founder and president of The Professor Is In, responded to the issue in [4] with "In the end we are all impoverished because one perpetrator often silences scores or hundreds of people whose scholarly contributions the world will never get to know," Kelsky said. "This is why I created the #MeTooPhD hashtag."

The NSF believes that academic institutions need to be more proactive in the way that they handle allegations of sexual misconduct. The NSF guidelines

`https://www.nsf.gov/news/news_summ.jsp?cntn_id=296610`

say

To improve accountability, NSF encourages anyone with a harassment complaint involving an NSF-funded researcher to report the incident to their institution and visit NSF's sexual harassment webpage.

`https://www.nsf.gov/od/odi/harassment.jsp`

From that site, individuals can submit harassment complaints directly to the NSF Office of Diversity and Inclusion. The webpage also includes "promising practices" on policies, effective codes of conduct and standards of behavior that may be applied everywhere NSF-supported research is conducted, as well as frequently asked questions.

You may not be comfortable reporting sexual harassment to your university and dealing with the investigation will take considerable time. This article [11] is a useful account of the process.

NIH has recently moved on this [29, 30] and has some on-line guidance

`https://www.nih.gov/anti-sexual-harassment`

Appendix A

Resources

A.1 ▪ Web Pages from Federal Agencies and Professional Societies

This list is far from complete, but has several good places to start your search for research support and networking opportunities.

- Federal Research Support
  - National Science Foundation:
    `https://www.nsf.gov`
  - US Department of Energy Advanced Scientific Computing Research:
    `https://www.energy.gov/science/ascr/advanced-scientific-computing-research`
  - US Army Research office:
    `https://arl.devcom.army.mil/who-we-are/aro`
- Professional Societies
  - Society for Industrial and Applied Mathematics: `https://www.siam.org`
  - Mathematical Association of America: `https://maa.org`
  - Americal Mathematical Society: `https://www.ams.org/home/page`

A.2 ▪ Working Life from Science Magazine

Science Magazine https://www.sciencemag.org is the flagship journal of the AAAS (Americal Association for the Advancement of Science). The magazine has a section in every issue called "Working Life". We have cited this many times in this book and used excerpts for some of the vignettes.

Here is an incomplete list of some recent articles from Science. Most of these are from "Working Life", but others cover things such as fraud and bullying.

Having children in graduate school	[19]
Being from outside the US	[44, 24, 55]
Bullying	[39, 43, 41, 34]
Dealing with setbacks	[6]
Depression	[1, 15, 12]
Financial problems	[45]
Fraud	[10, 38, 37, 35, 47]
Getting fair credit for your work	[16, 18, 5]
Giving talks	[27]
Harassment	[29, 21]
Positive thinking	[28]
Underrepresented groups/bias	[46, 40, 49, 13, 2]
Work-life balance	[25, 56, 53, 8]

Bibliography

[1] S. M. AGUILLON, *Fighting through the gray*, Science, 363 (2019), p. 1114, doi:10.1126/science.363.6431.1114, https://science.sciencemag.org/content/363/6431/1114. (Cited on pp. 25, 70)

[2] ANONYMOUS, *The things I wish I could say*, Science, 370 (2020), p. 630. (Cited on pp. 66, 70)

[3] Y. BHATTACHARJEE, *The mind of a con man*, New York Times, (April 6, 2013), https://www.nytimes.com/2013/04/28/magazine/diederik-stapels-audacious-academic-fraud.html. (Cited on p. 65)

[4] R. BIHN-WALLACE, *Yuval peres, math professor with series of sexual misconduct allegations levied against him, gives lecture at UC Davis*, The California Aggie, (December 5, 2019). (Cited on p. 67)

[5] K. BISHT, *A ghost author speaks up*, Science, 373 (2021), p. 1274. (Cited on pp. 19, 70)

[6] M. BJÖRNMALM, *Taking a break is hard work, too*, Science, 364 (2019), p. 98, doi:10.1126/science.364.6435.98. (Cited on p. 70)

[7] K. M. BLISS, B. J. GALLUZZO, K. R. KAVANAGH, AND R. LEVY, *Math Modeling: Computing and Communicating*, SIAM, Philadelphia, 2018. (Cited on p. 21)

[8] A. M. BOURKE, *Waking up to my sleep disorder*, Science, 373 (2021), p. 1054, doi:10.1126/science.373.6558.1054. (Cited on p. 70)

[9] K. BREDBENNER, *One Ph.D., hold the pastries*, Science, 365 (2019), p. 94, doi:DOI:10.1126/science.365.6448.94. (Cited on p. 11)

[10] C. O'GRADY, *Honesty study was based on fabricated data*, Science, 373 (2021), pp. 950–951, doi:10.1126/science.373.6558.950. (Cited on p. 70)

[11] S. A. COLLINS KEY, *After my Ph.D. adviser was investigated for sexual misconduct, it took a community to help me recover*, Science, 375 (2022), p. 114, doi:10.1126/science.caredit.acz9951. (Cited on p. 68)

[12] E. A. D., *Out of the fog*, Science, 374 (2021), p. 902, https://science.sciencemag.org/content/374/6569/902. (Cited on pp. 25, 70)

[13] L. T. DAS, *When the jokes aren't funny*, Science, 369 (2020), p. 342. (Cited on pp. 66, 70)

[14] D. DICKERSON, *Strange dreams*, Science, 367 (2020), p. 114, https://science.sciencemag.org/content/367/6473/114. (Cited on p. 15)

[15] L. DILKES-HOFFMAN, *When the trail gets steep*, Science, 363 (2019), p. 662, doi:10.1126/science.363.6427.662, https://science.sciencemag.org/content/363/6427/662. (Cited on pp. 25, 70)

[16] M. M. ELMASSRY, *Helping others - and myself*, Science, 368 (2020), p. 1282. (Cited on pp. 19, 70)

[17] H. ELSE, *Does science have a bullying problem?*, Nature, 563 (2018), pp. 616–618, https://www.nature.com/articles/d41586-018-07532-5. (Cited on p. 65)

[18] E. FOGARTY, *Credit where credit is due*, Science, 370 (2020), p. 1130, https://www.science.org/doi/10.1126/science.370.6520.1130. (Cited on pp. 19, 70)

[19] R. GALTON, *The parent trap*, Science, 365 (2019), p. 1214, doi:10.1126/science.365.6458.1214. (Cited on p. 70)

[20] V. GEWIN, *Can Germany rein in its academic bullying problem*, Nature, 641 (2025), pp. 545–547, https://www.nature.com/articles/d41586-025-01207-8. (Cited on p. 65)

[21] A. GIBBONS, *Prominent Harvard anthropologist put on leave*, Science, 368 (2020), pp. 1172–1173. (Cited on pp. 67, 70)

[22] G. D. GOPEN, *Getting the point across*, American Scientist, 110 (2022), pp. 346–351. (Cited on p. 16)

[23] G. D. GOPEN, *Writing to facilitate reading*, American Scientist, 111 (2023), pp. 28–33. (Cited on p. 16)

[24] J. HARIHARAN, *Uncovering the hidden curriculum*, Science, 364 (2019), p. 702, doi:10.1126/science.364.6441.702. (Cited on p. 70)

[25] M. Z. HE, *Singing for science*, Science, 364 (2019), p. 506, doi:10.1126/science.364.6439.506, https://science.sciencemag.org/content/364/6439/506. (Cited on p. 70)

[26] N. J. HIGHAM, *Handbook of Writing for the Mathematical Sciences*, Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, third ed., 2020. (Cited on pp. 16, 21, 38, 56)

[27] M. HIS, *Stand out as a speaker*, Science, 365 (2019), p. 834, doi:10.1126/science.365.6455.834. (Cited on p. 70)

[28] S. HOLZER, *Don't panic*, Science, 376 (2022), p. 886. (Cited on p. 70)

[29] J. KAISER, *NIH requires disclosure about sexual harassment by grantees*, Science, 368 (2020), p. 1301. (Cited on pp. 63, 68, 70)

[30] J. KAISER, *NIH rolls out stricter disclosure rules for sexual harrassment*, Science, 376 (2022), p. 787. (Cited on pp. 63, 68)

[31] T. KOLDA AND V. TORCZON, *Top ten ways to lose an audience*, SIAM News, 44 (2011), https://archive.siam.org/pdf/news/1876.pdf. (Cited on p. 47)

[32] *Better letters: Equitable practices for writing, reading and soliciting letters of recommendation.* https://sites.google.com/igenetwork.org/equity-in-graduate-education/home/resourceguide2, 2020. (Cited on p. 47)

[33] R. LEVY, R. LAUGESEN, AND F. SANTOSA, *BIG Jobs Guide: Business, Industry, and Government Careers for Mathematical Scientists, Statisticians, and Operations Researchers*, SIAM, Philadelphia, 2018. (Cited on pp. 56, 57)

[34] M. MAHMOUDI, *Academic bullying: Desperate for data and solutions*, 2020, https://www.sciencemag.org/features/2020/01/academic-bullying-desperate-data-and-solutions. (Cited on pp. 64, 70)

[35] E. MARSHALL, *Fraud strikes to genome lab*, 1996. (Cited on pp. 66, 70)

[36] MATHEMATICAL ASSOCIATION OF AMERICA, *A handbook for mathematics teaching assistants*. https://www.maa.org/programs/students/student-resources/a-handbook-for-mathematics-teaching-assistants, 1999. (Cited on p. 17)

[37] J. MERVIS, *Must-pass U.S. defense bill would harmonize funding disclosures for all federal grant applicants*, doi:doi:10.1126/science.abg0171. (Cited on pp. 66, 70)

[38] J. MERVIS, *Paperwork mistakes ensnare health data expert mismanaging an NSF grant*, Science, 368 (2020), pp. 567–568, https://science.sciencemag.org/content/368/6491/567. (Cited on pp. 65, 66, 70)

[39] J. MERVIS AND M. STELLINO, *Hard lessons*, Science, 368 (2020), pp. 814–817, https://science.sciencemag.org/content/368/6493/814. (Cited on pp. 65, 70)

[40] B. MUSTANSKI, *Keep quiet about homophobia or open up?*, Science, 373 (2021), p. 826, doi:DOI:10.1126/science.373.6556.826. (Cited on p. 70)

[41] A. NAIK, *Choose your adviser wisely*, Science, 369 (2020), p. 1026, https://science.sciencemag.org/content/369/6506/1026. (Cited on pp. 64, 65, 70)

[42] OPENAI, *Chatgpt (mar 14 version)*, 2024, https://chat.openai.com. Accessed: 2025-05-11. (Cited on p. 6)

[43] R. POOLE, *Bullied out of research*, Science, 354 (2016), p. 514. (Cited on pp. 64, 70)

[44] L. A. RODRIGUEZ-CRUZ, *Between two worlds*, Science, 365 (2019), p. 946, doi:10.1126/science.365.6456.946. (Cited on p. 70)

[45] J. E. SAGERS, *Waiting for reimbursement*, Science, 363 (2019), p. 786, doi:10.1126/science.363.6428.786. (Cited on pp. 26, 70)

[46] S. SANTOS-DÍAZ, *When support becomes pressure*, Science, 363 (2019), p. 1006, doi:10.1126/science.363.6430.1006. (Cited on p. 70)

[47] SCIENCE NEWS STAFF, *United States charges prominent Harvard chemist with failing to disclose China ties*, 2020, https://www.sciencemag.org/news/2020/01/us-charges-prominent-harvard-chemist-failing-disclose-china-ties. (Cited on pp. 59, 66, 70)

[48] E. SKARAKIS-DOYLE AND G. G. MCINTYRE, *Western Guide to Graduate Supervision*, U. Western Ontario, London, Ontario, Canada, 2008. (Cited on p. 36)

[49] M. SPIKES, *The pressure to assimilate*, Science, 368 (2020), p. 1506. (Cited on pp. 66, 70)

[50] E. R. TUFTE, *The Visual Display of Quantative Information*, Graphics Press, Cheshire, CT, 2001. (Cited on p. 16)

[51] A. TVEITO, *On supervising PhD students*, https://www.simula.no/sites/default/files/supervision_by_aslak_tveito.pdf. (Cited on p. 36)

[52] M. WADMAN, *NIH stumbles on 'pass the harasser'*, Science, 390 (2023), pp. 441–443. (Cited on p. 63)

[53] E. R. WENGERT, *Learning to unplug*, Science, 373 (2021), p. 938, doi:10.1126/science.373.6557.938. (Cited on p. 70)

[54] S. WEST AND L. TURNBULL, *Scientific Papers Made Easy: How to Write with Clarity and Impact in the Life Sciences*, Oxford University Press, Oxford, 2023. (Cited on p. 16)

[55] J. E. L. Y GONZÁLEZ, *We regret to inform you*, Science, 364 (2019), p. 302, doi:10.1126/science.364.6437.302. (Cited on p. 70)

[56] L. YONA, *A scientist and an advocate*, Science, 363 (2019), p. 1474, doi:10.1126/science.363.6434.1474, https://science.sciencemag.org/content/363/6434/1474. (Cited on p. 70)

[57] M. ZUO, *Top Chinese university sacks academic accused of inappropriate relationships with string of students*. https://www.scmp.com/news/china/society/article/3041818/top-chinese-university-sacks-academic-accused-dating-string, 2019. (Cited on p. 67)

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